

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

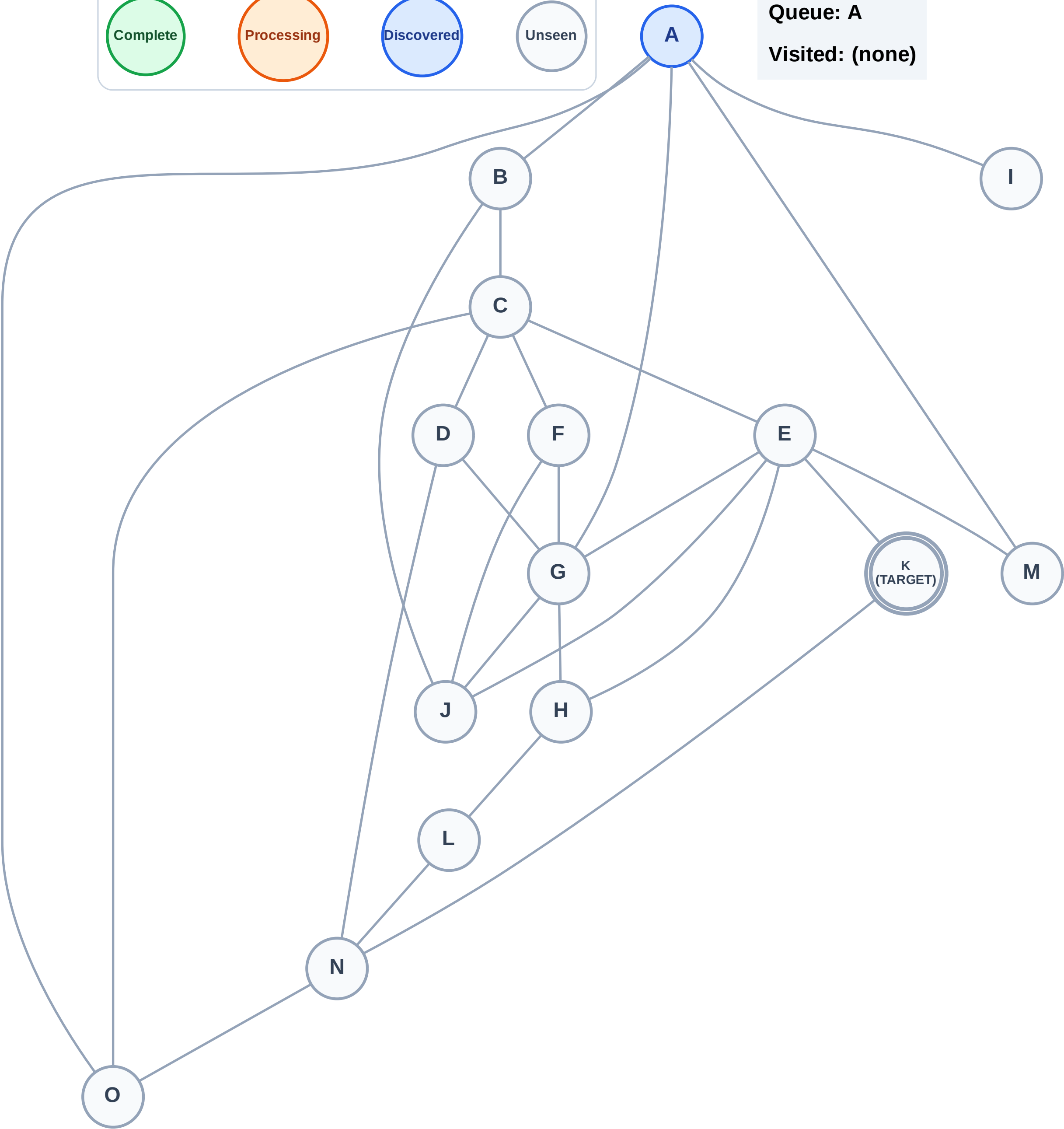
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

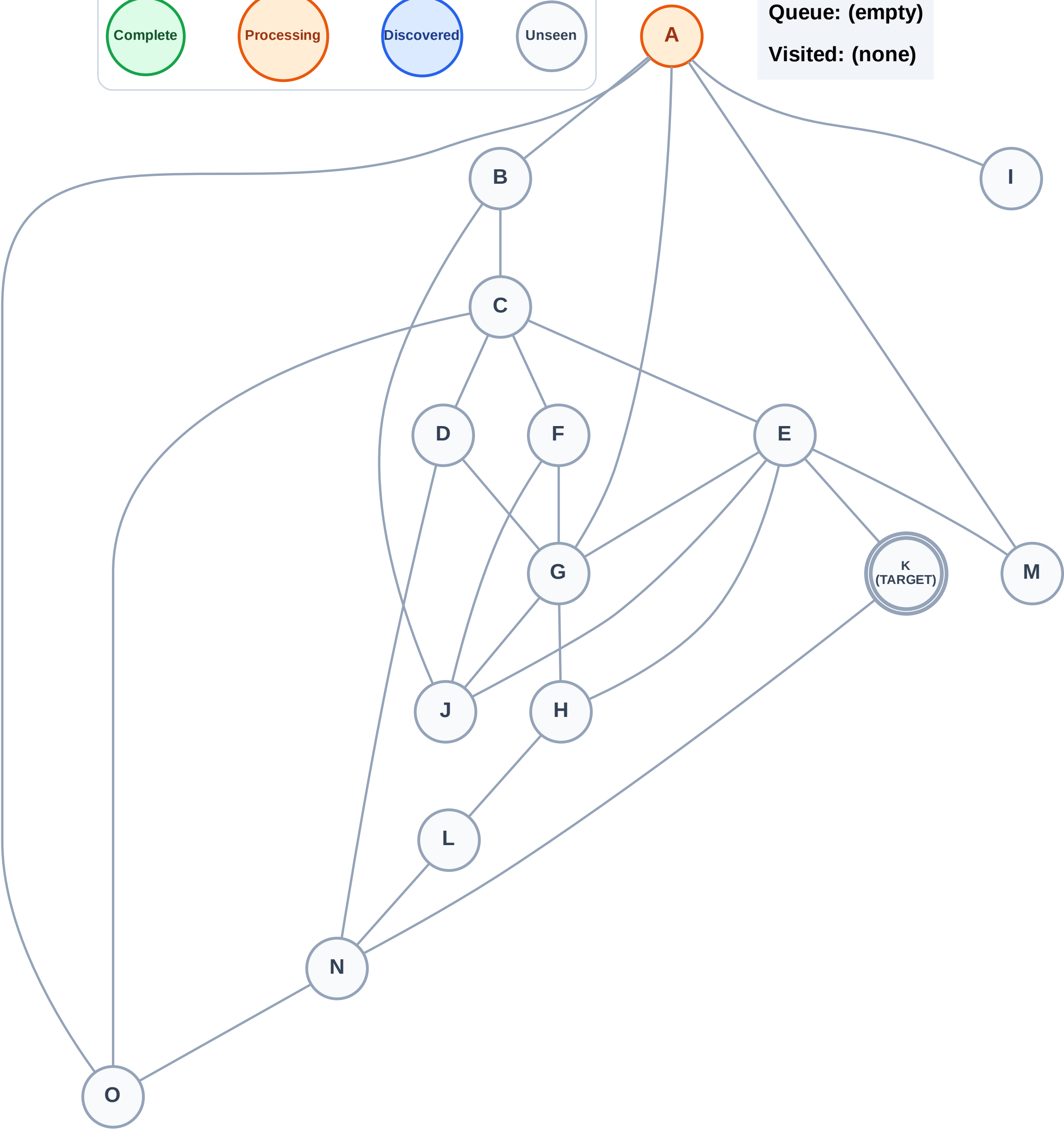
Complete

Processing

Discovered

Unseen

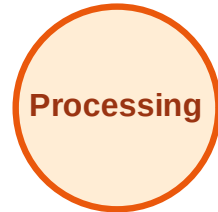
Queue: (empty)
Visited: (none)



BFS Traversal

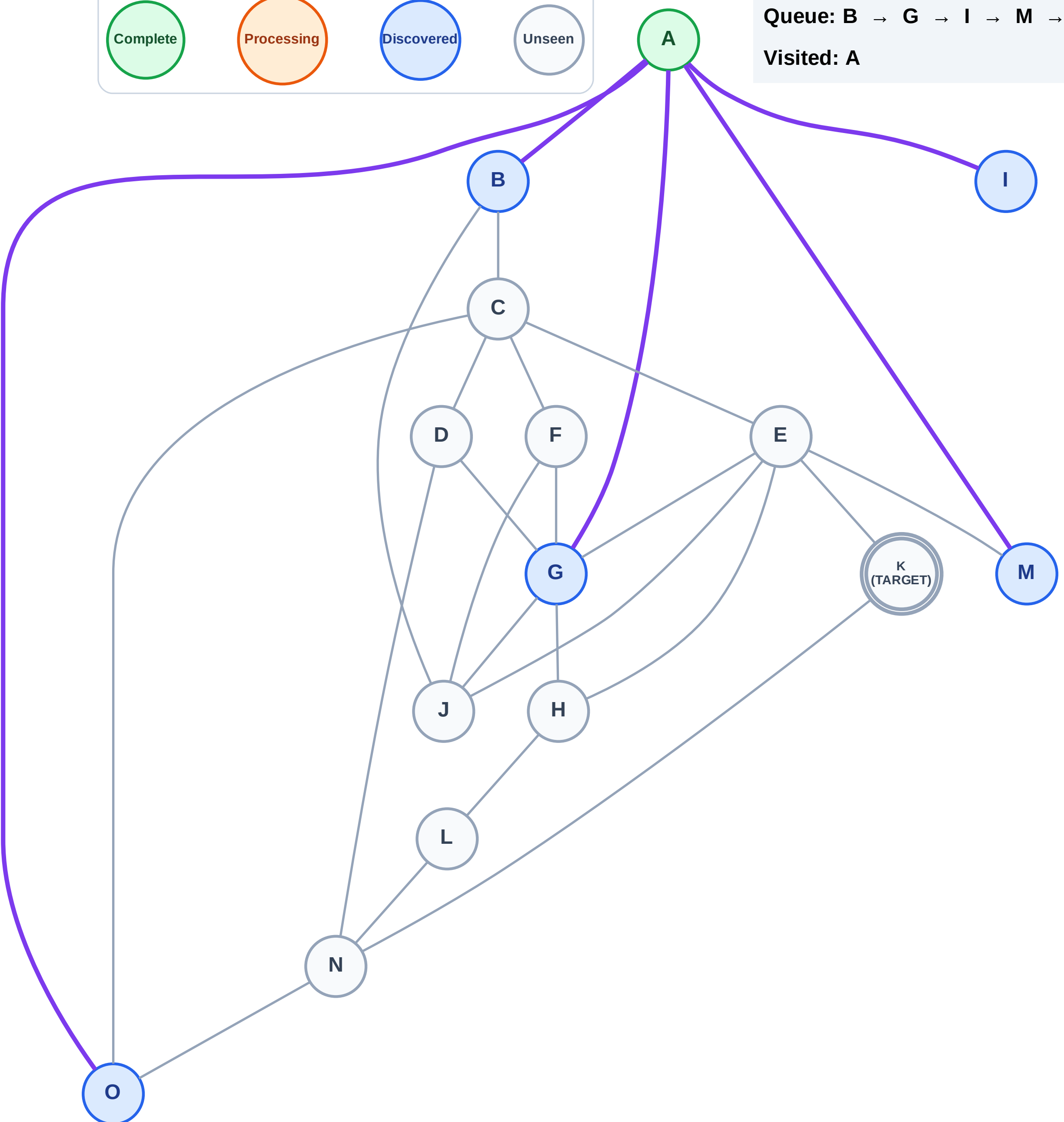
Step 3: Discover B, G, I, M, O from A

Legend



Queue: B → G → I → M → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

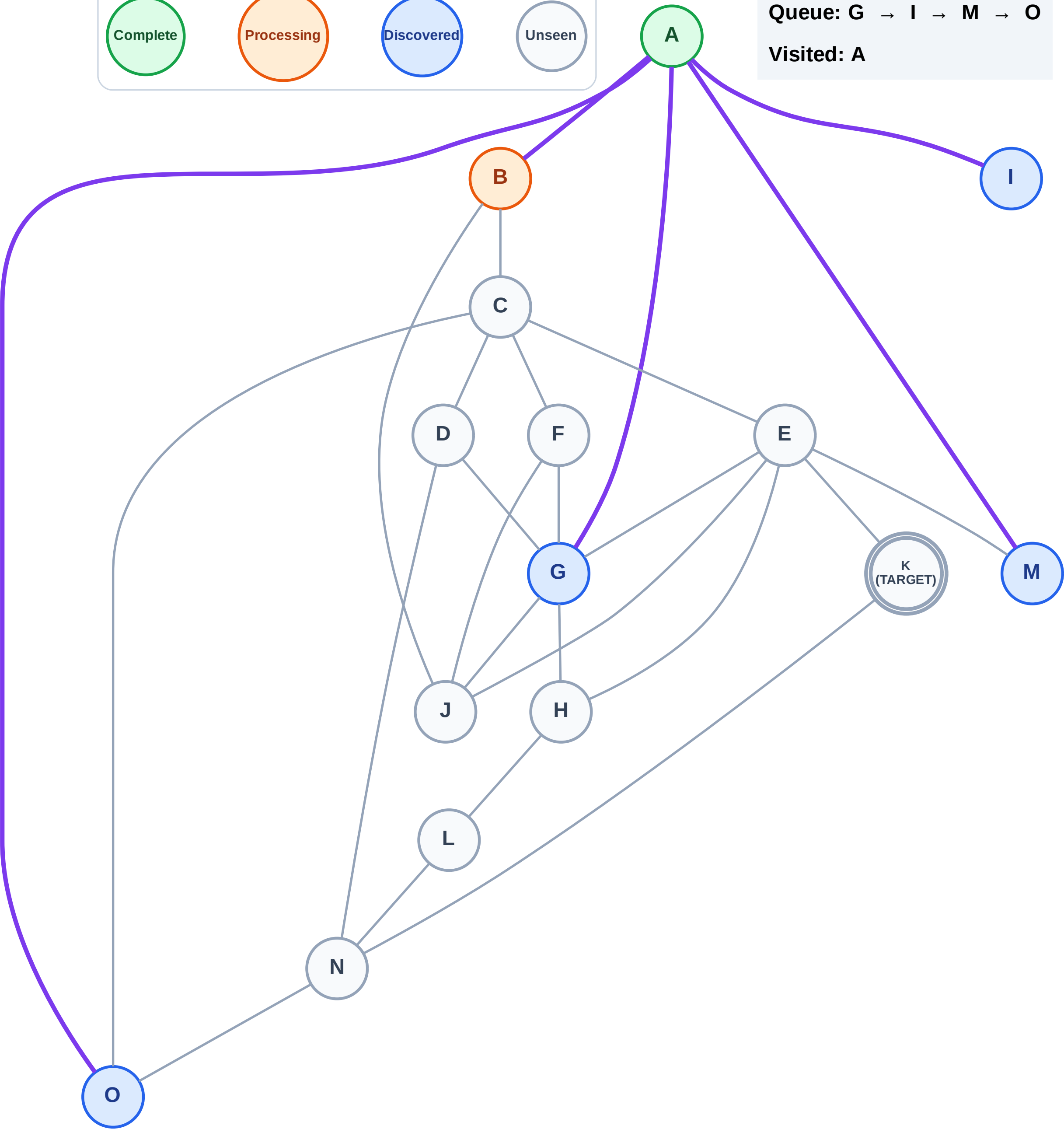
Processing

Discovered

Unseen

Queue: G → I → M → O

Visited: A



BFS Traversal

Step 5: Discover C, J from B

Legend

Complete

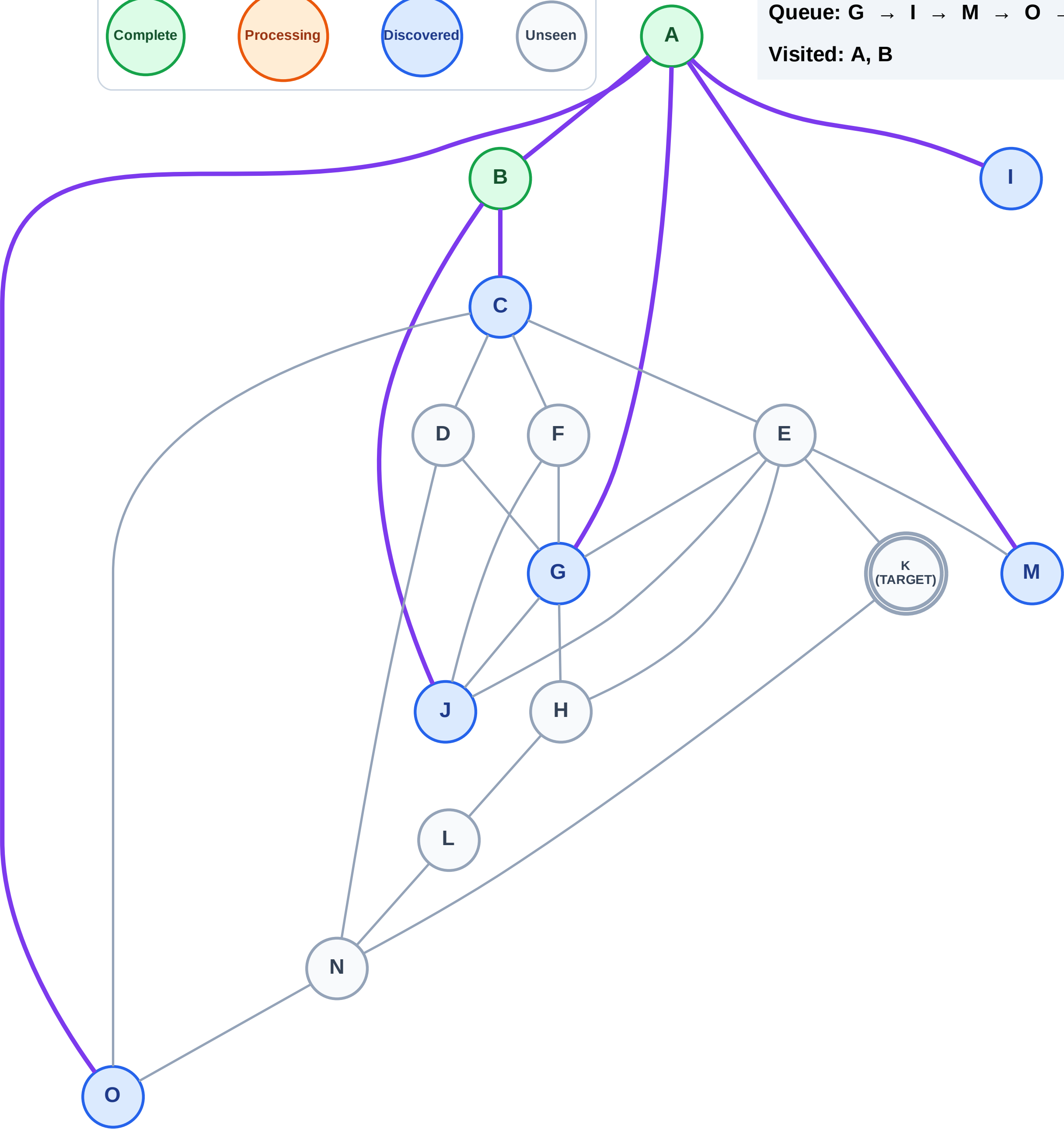
Processing

Discovered

Unseen

Queue: G → I → M → O → C → J

Visited: A, B



BFS Traversal

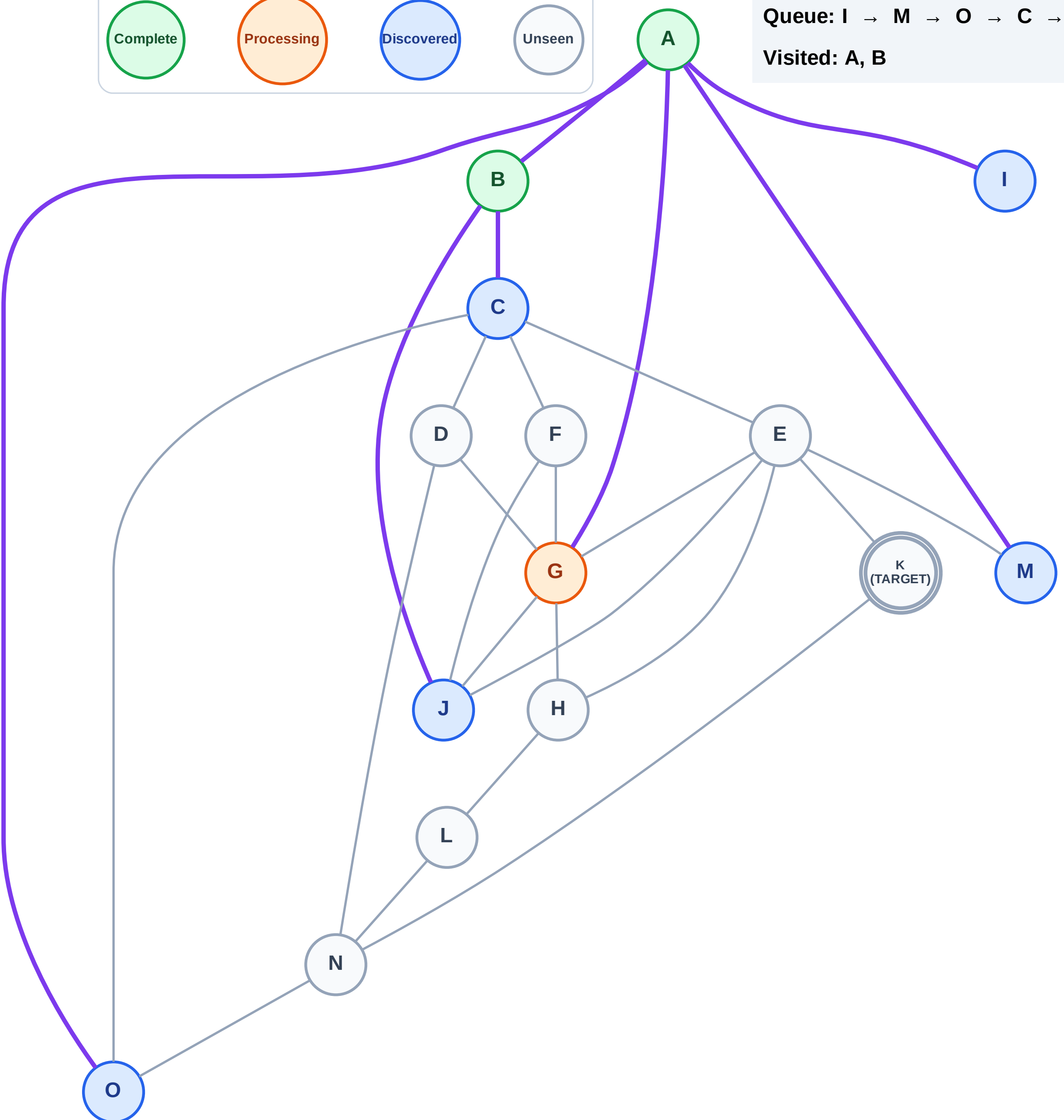
Step 6: Process node G

Legend



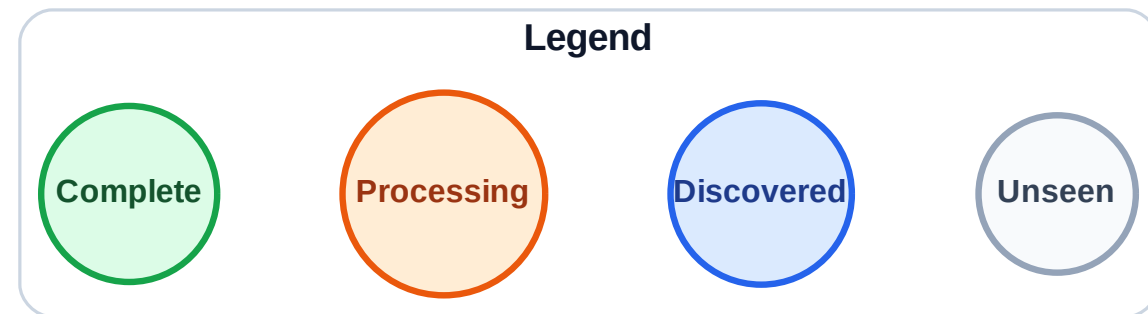
Queue: I → M → O → C → J

Visited: A, B



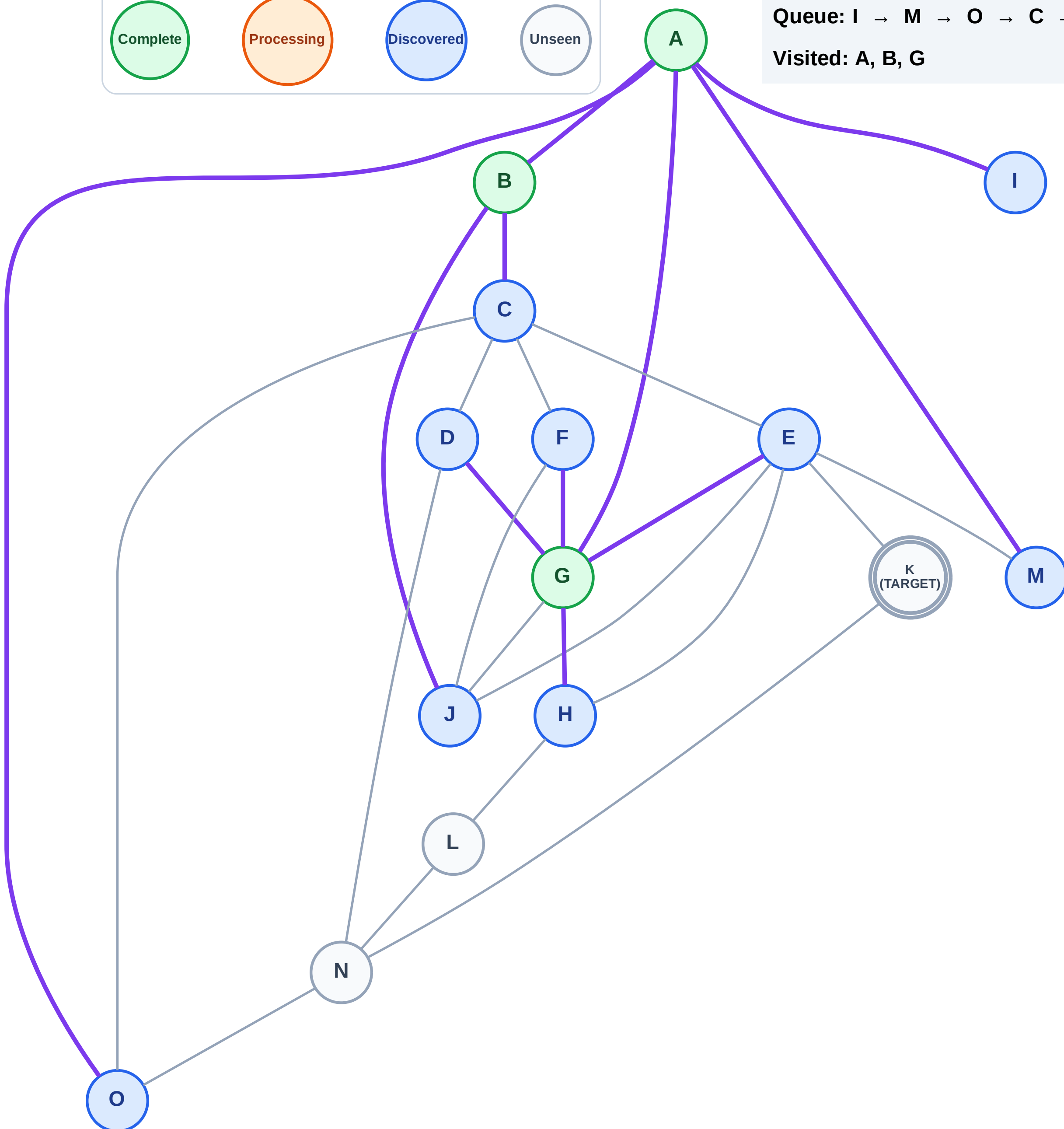
BFS Traversal

Step 7: Discover D, E, F, H from G



Queue: I → M → O → C → J → D → E → F → H

Visited: A, B, G



BFS Traversal

Step 8: Process node I

Legend

Complete

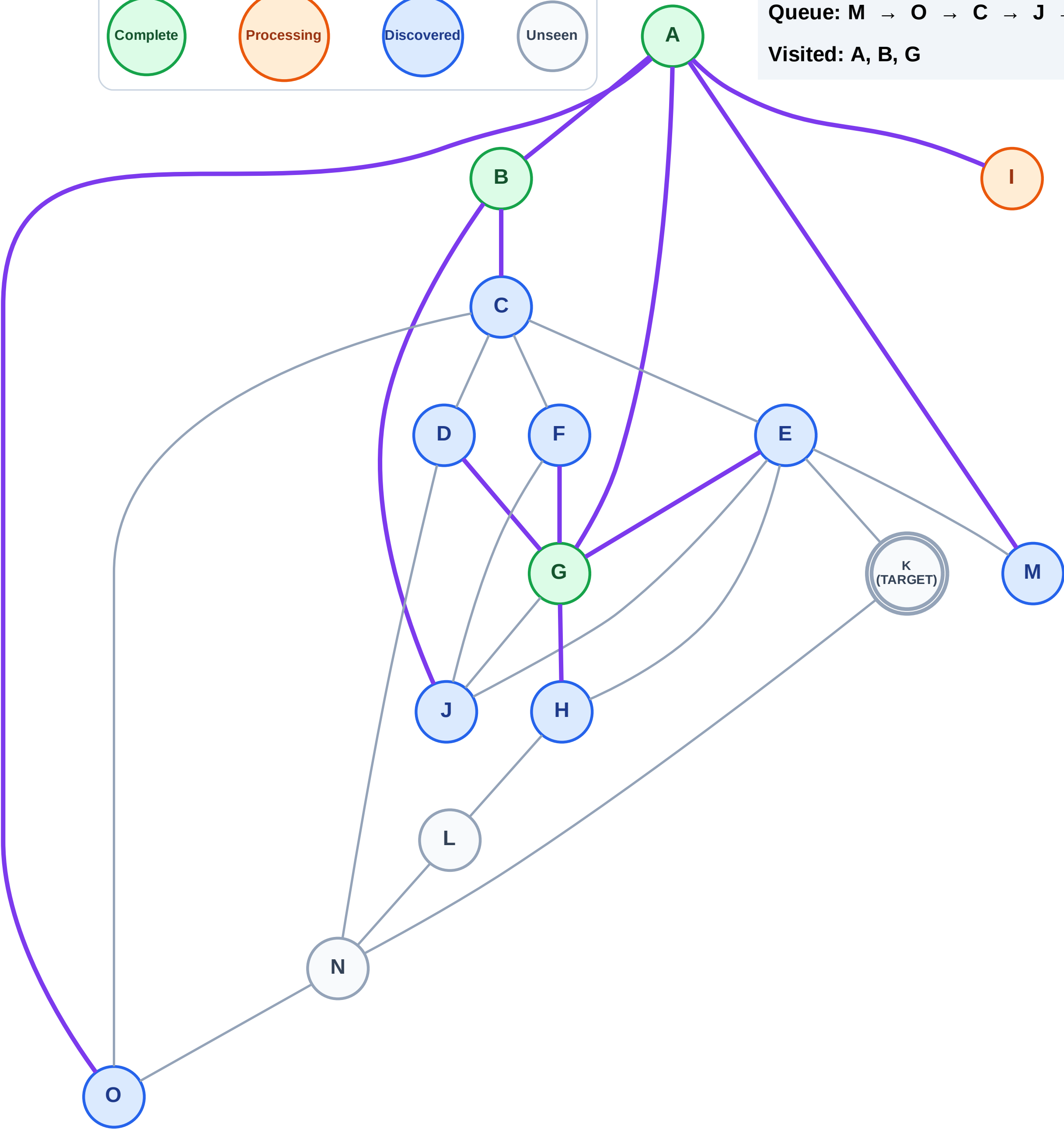
Processing

Discovered

Unseen

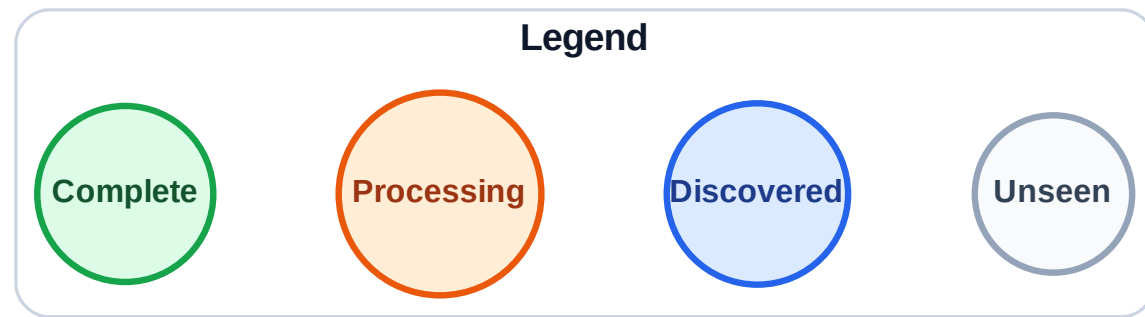
Queue: M → O → C → J → D → E → F → H

Visited: A, B, G

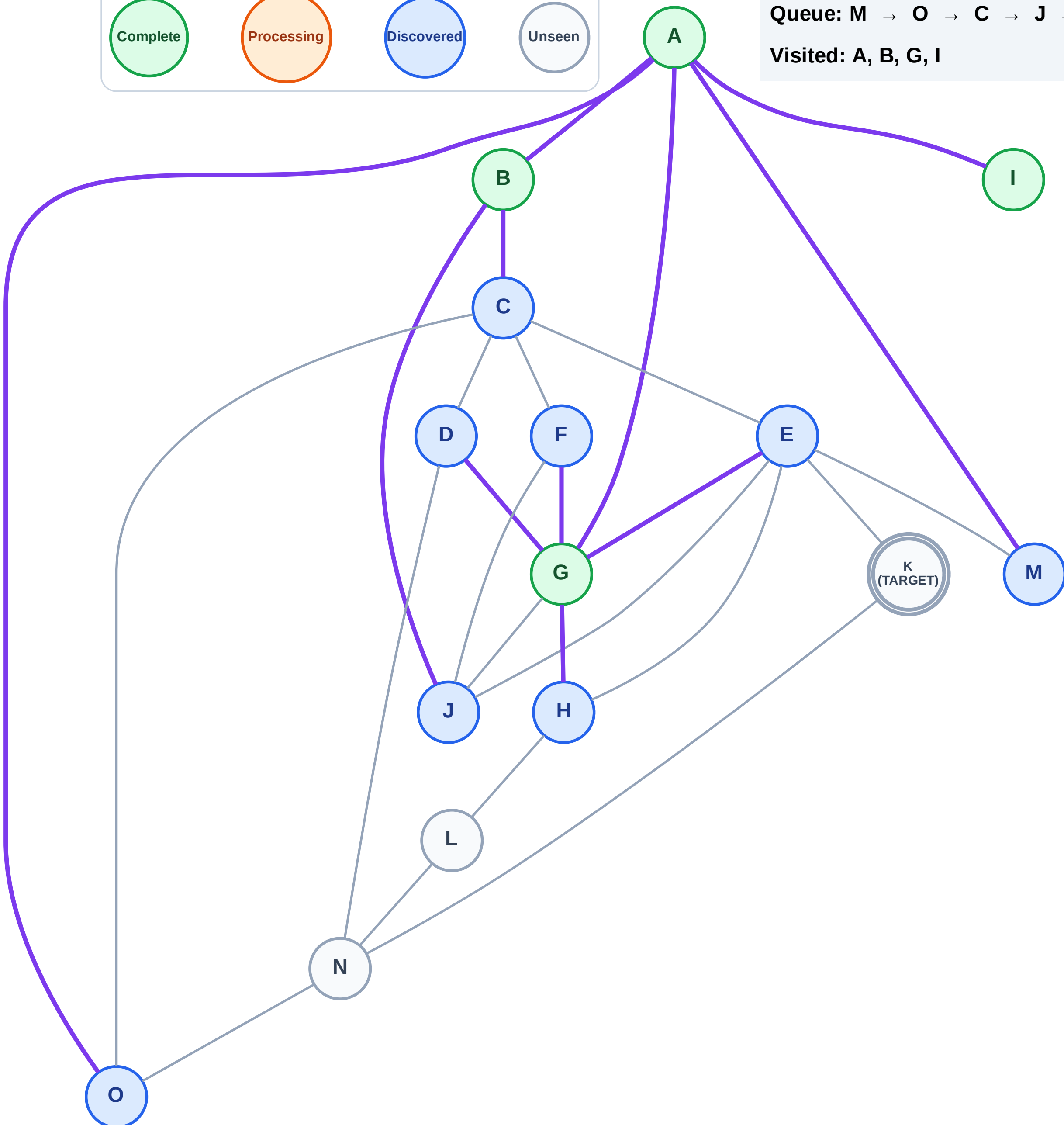


BFS Traversal

Step 9: No new neighbors from I



Queue: M → O → C → J → D → E → F → H
Visited: A, B, G, I



BFS Traversal

Step 10: Process node M

Legend

Complete

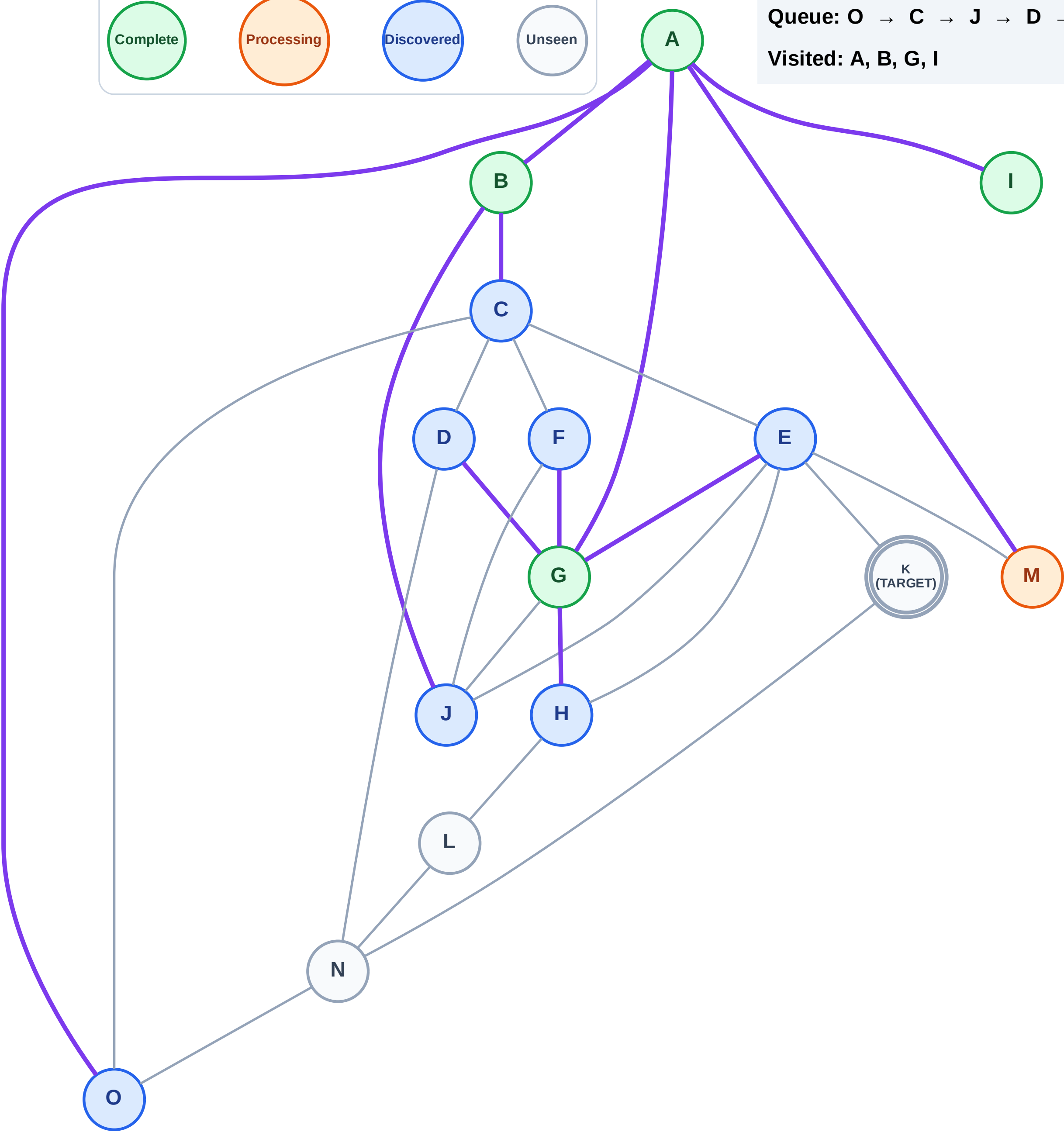
Processing

Discovered

Unseen

Queue: O → C → J → D → E → F → H

Visited: A, B, G, I



BFS Traversal

Step 11: No new neighbors from M

Legend

Complete

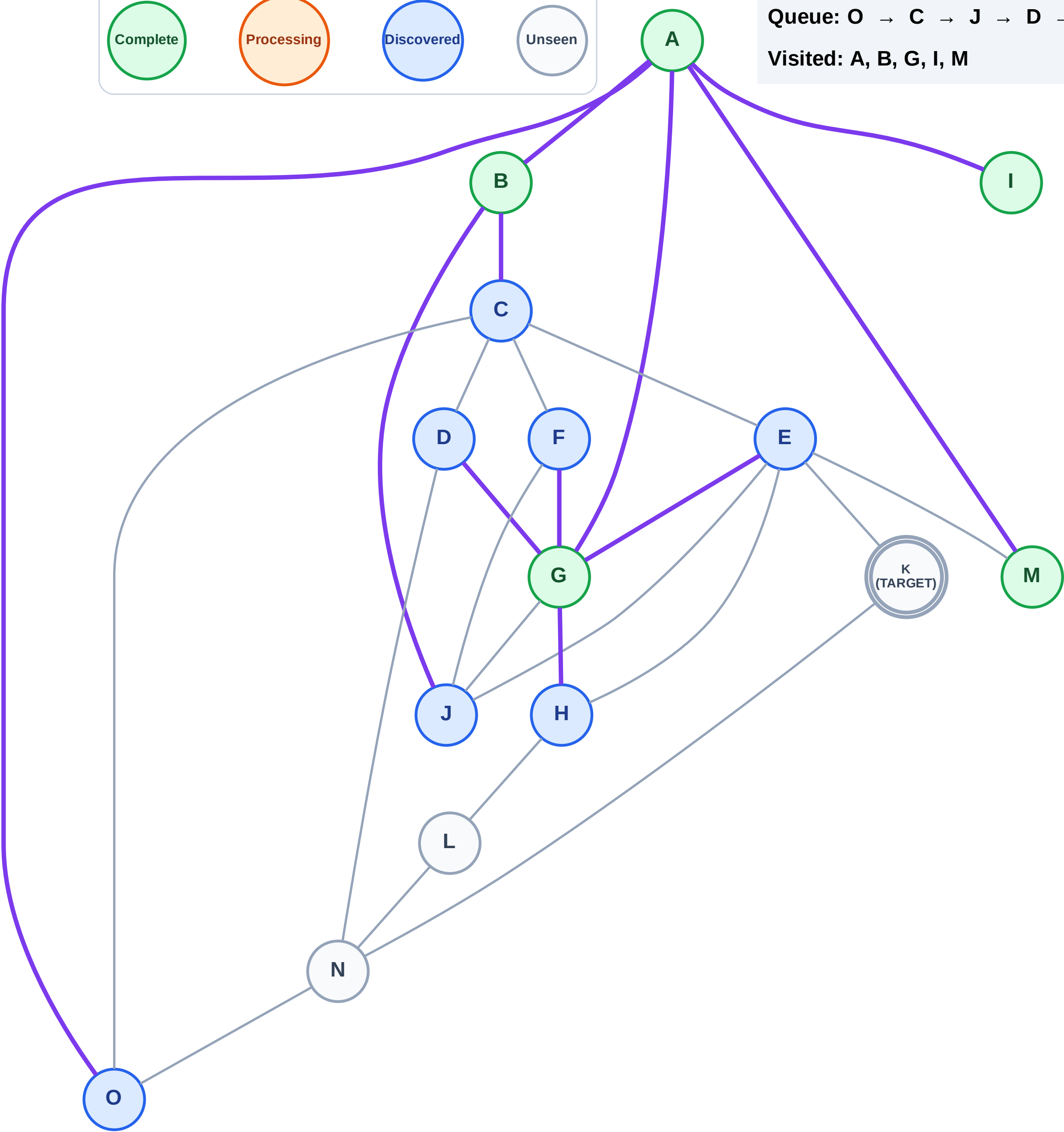
Processing

Discovered

Unseen

Queue: O → C → J → D → E → F → H

Visited: A, B, G, I, M



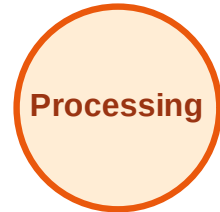
BFS Traversal

Step 12: Process node O

Legend



Complete



Processing



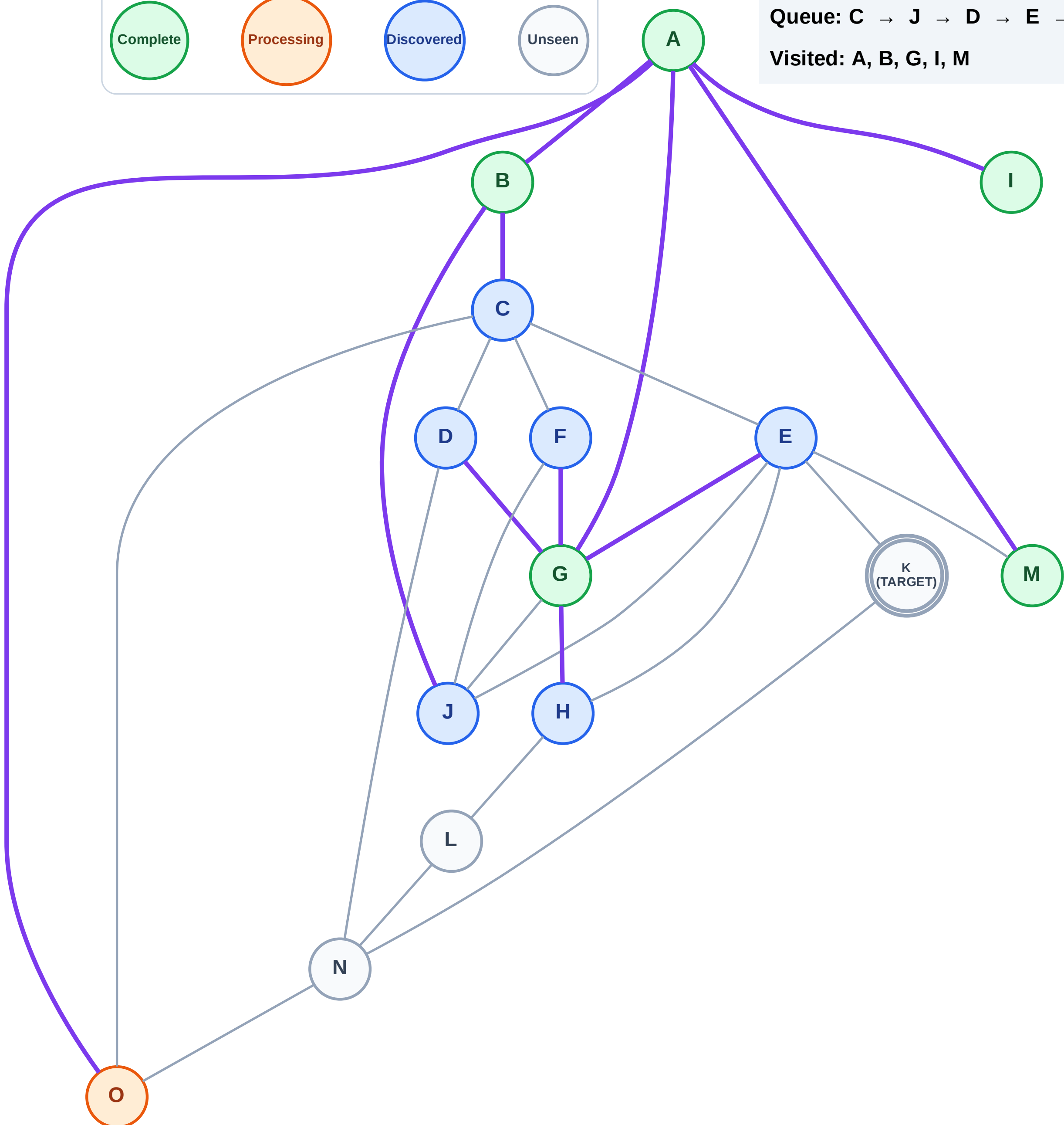
Discovered



Unseen

Queue: C → J → D → E → F → H

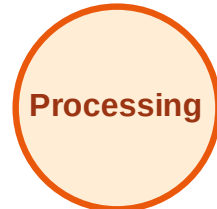
Visited: A, B, G, I, M



BFS Traversal

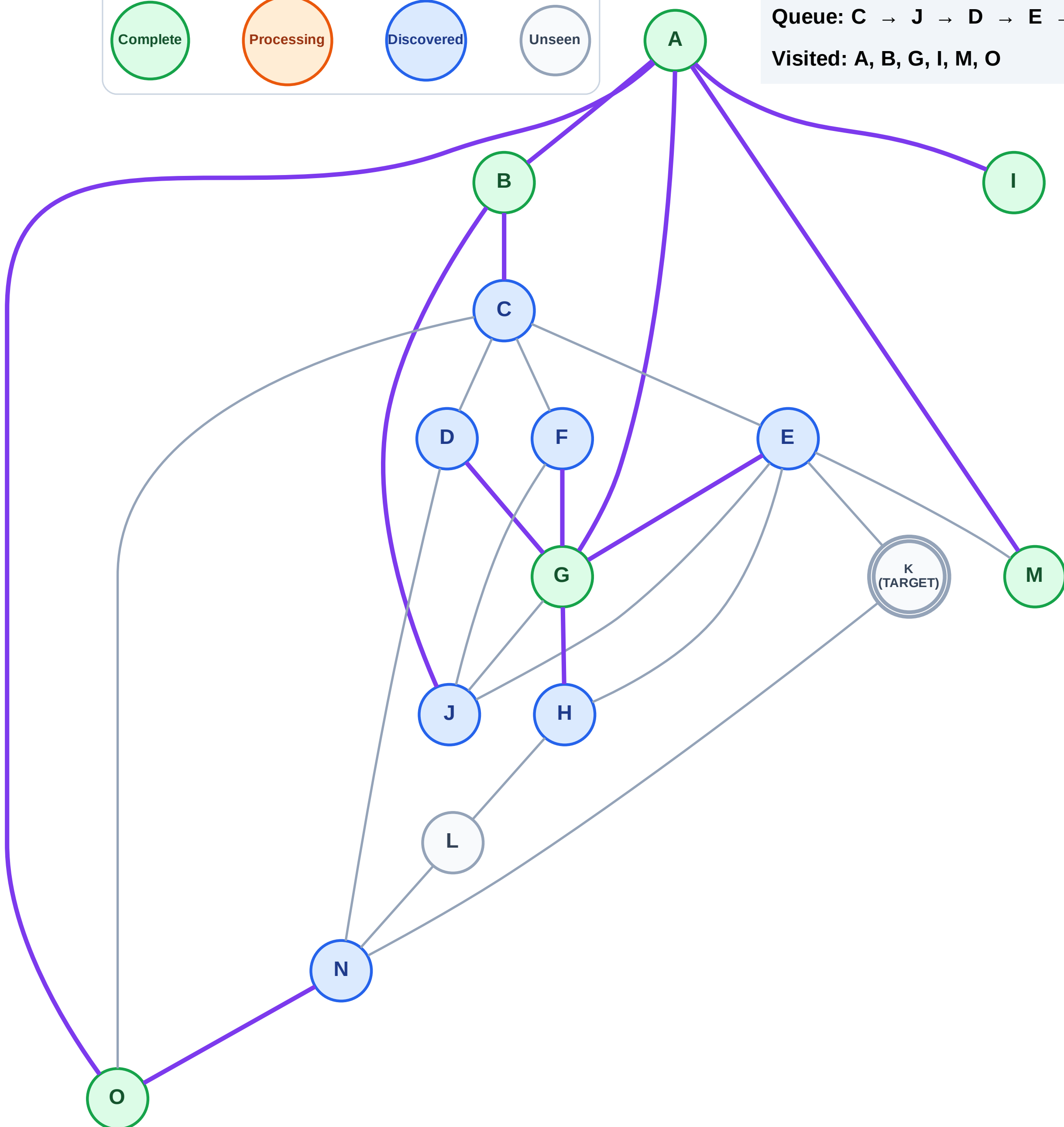
Step 13: Discover N from O

Legend



Queue: C → J → D → E → F → H → N

Visited: A, B, G, I, M, O



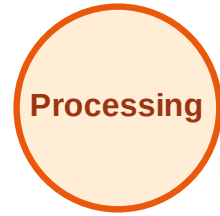
BFS Traversal

Step 14: Process node C

Legend



Complete



Processing



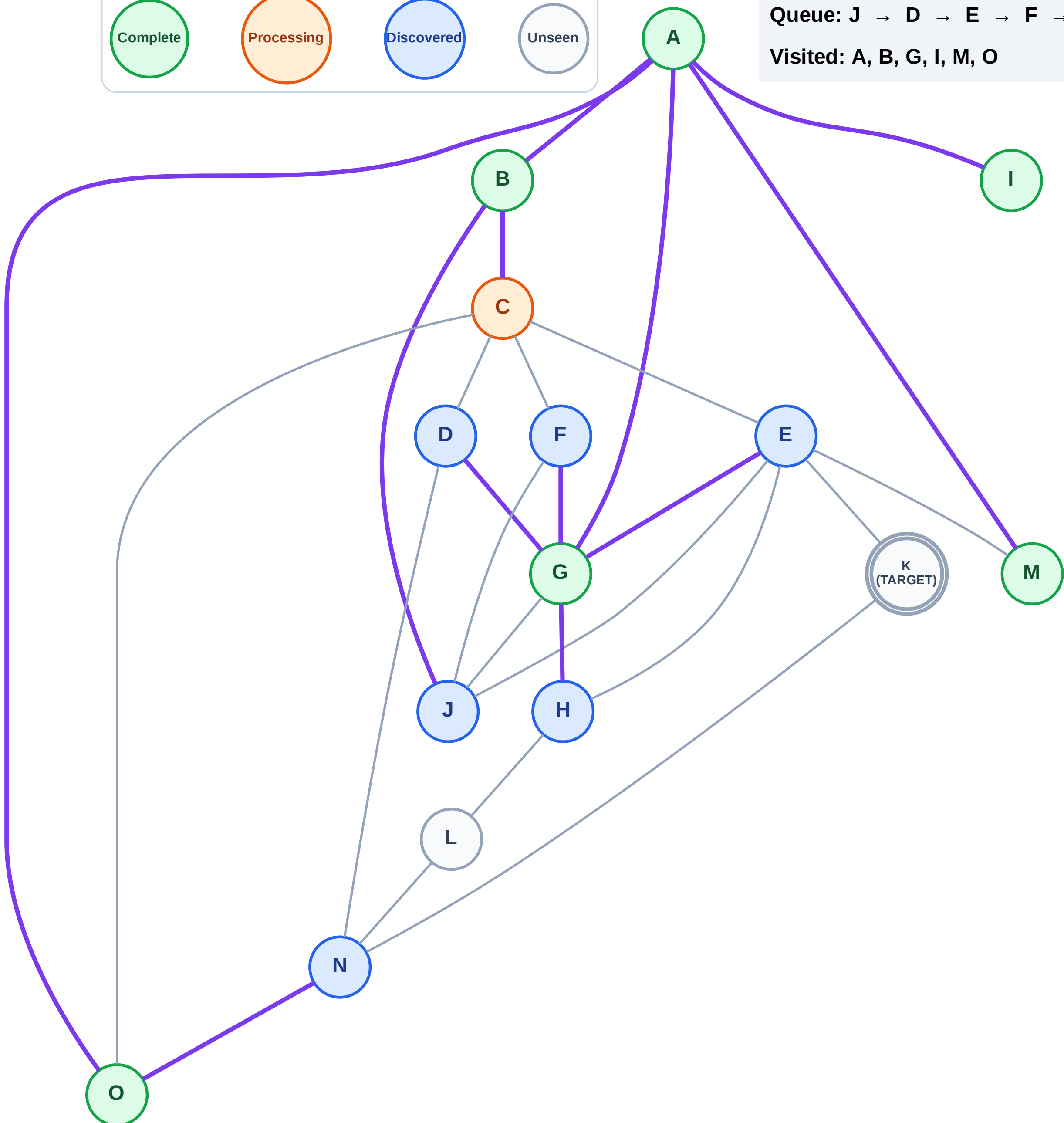
Discovered



Unseen

Queue: J → D → E → F → H → N

Visited: A, B, G, I, M, O



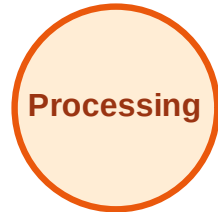
BFS Traversal

Step 15: No new neighbors from C

Legend



Complete



Processing



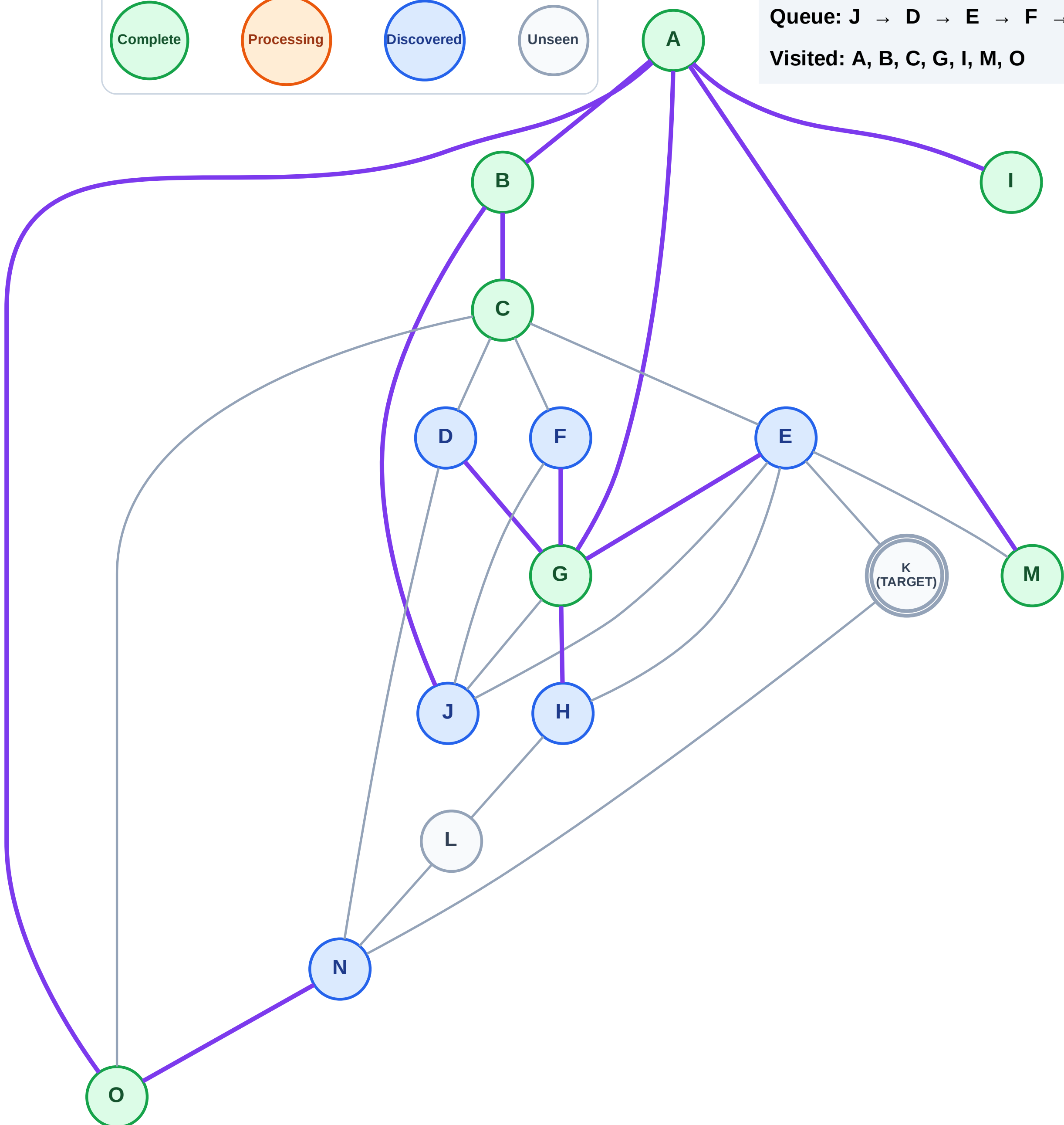
Discovered



Unseen

Queue: J → D → E → F → H → N

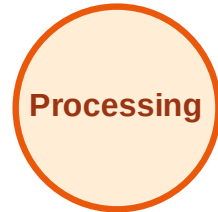
Visited: A, B, C, G, I, M, O



BFS Traversal

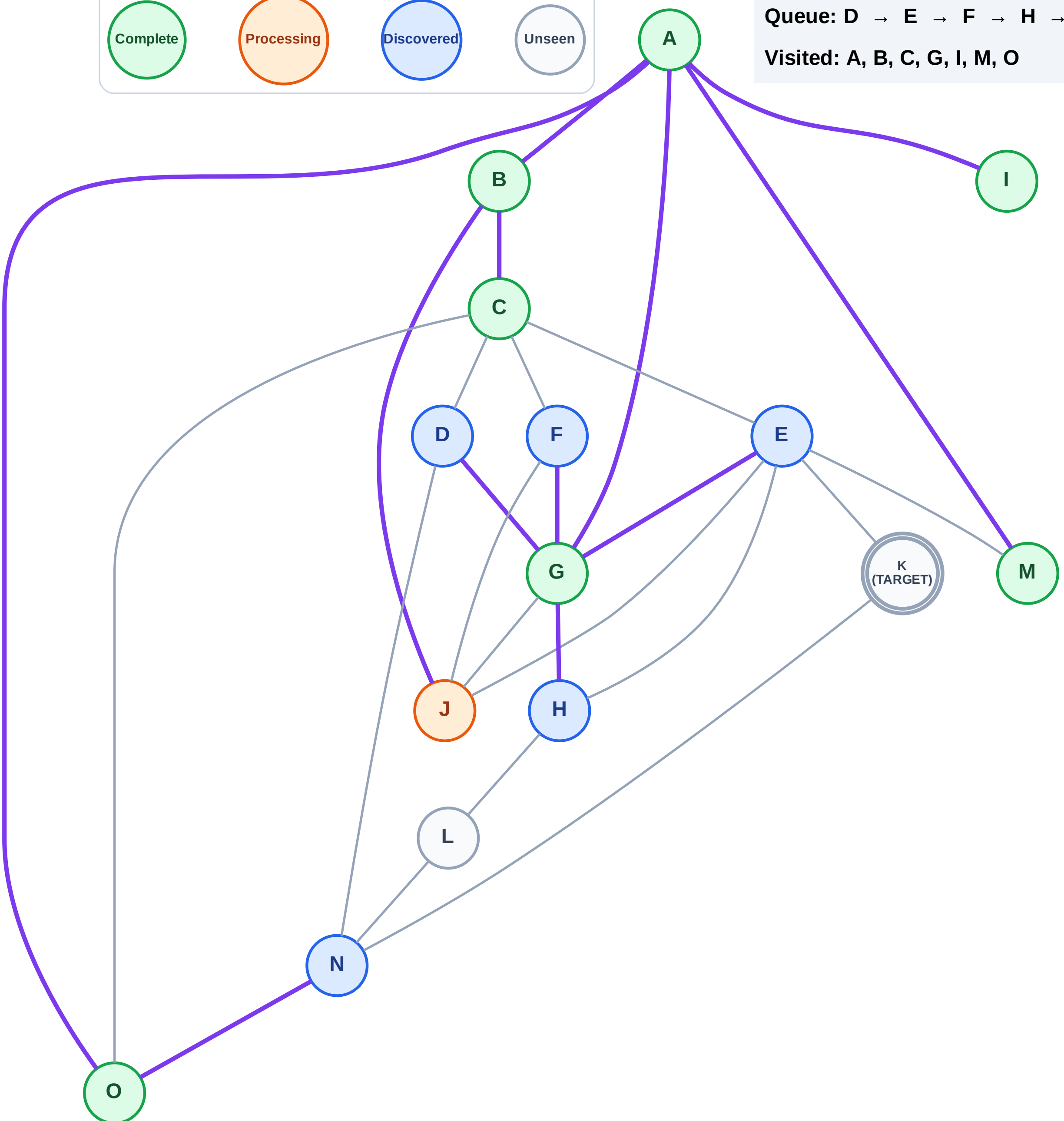
Step 16: Process node J

Legend



Queue: D → E → F → H → N

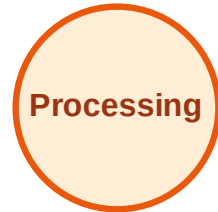
Visited: A, B, C, G, I, M, O



BFS Traversal

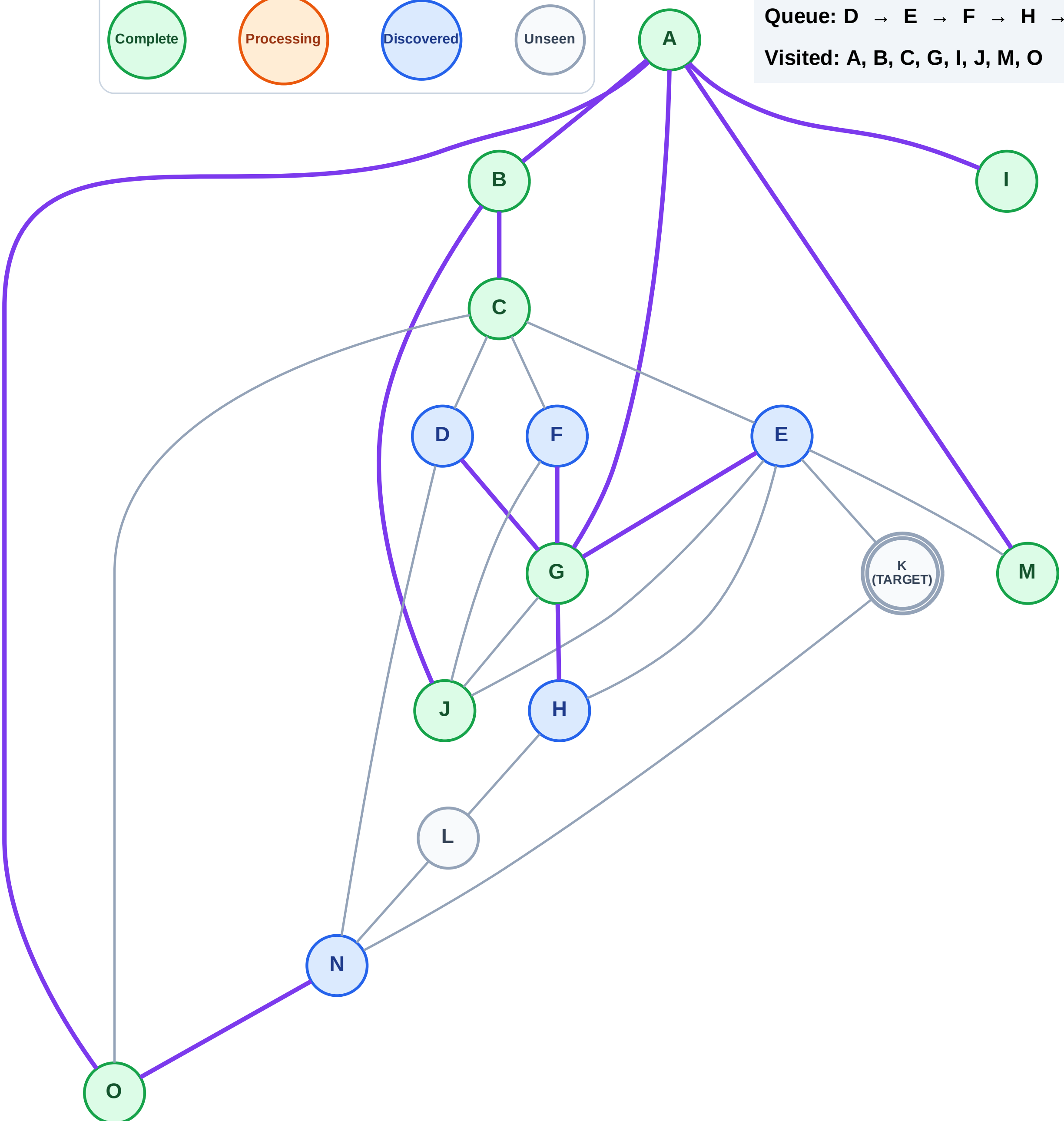
Step 17: No new neighbors from J

Legend



Queue: D → E → F → H → N

Visited: A, B, C, G, I, J, M, O



BFS Traversal

Step 18: Process node D

Legend

Complete

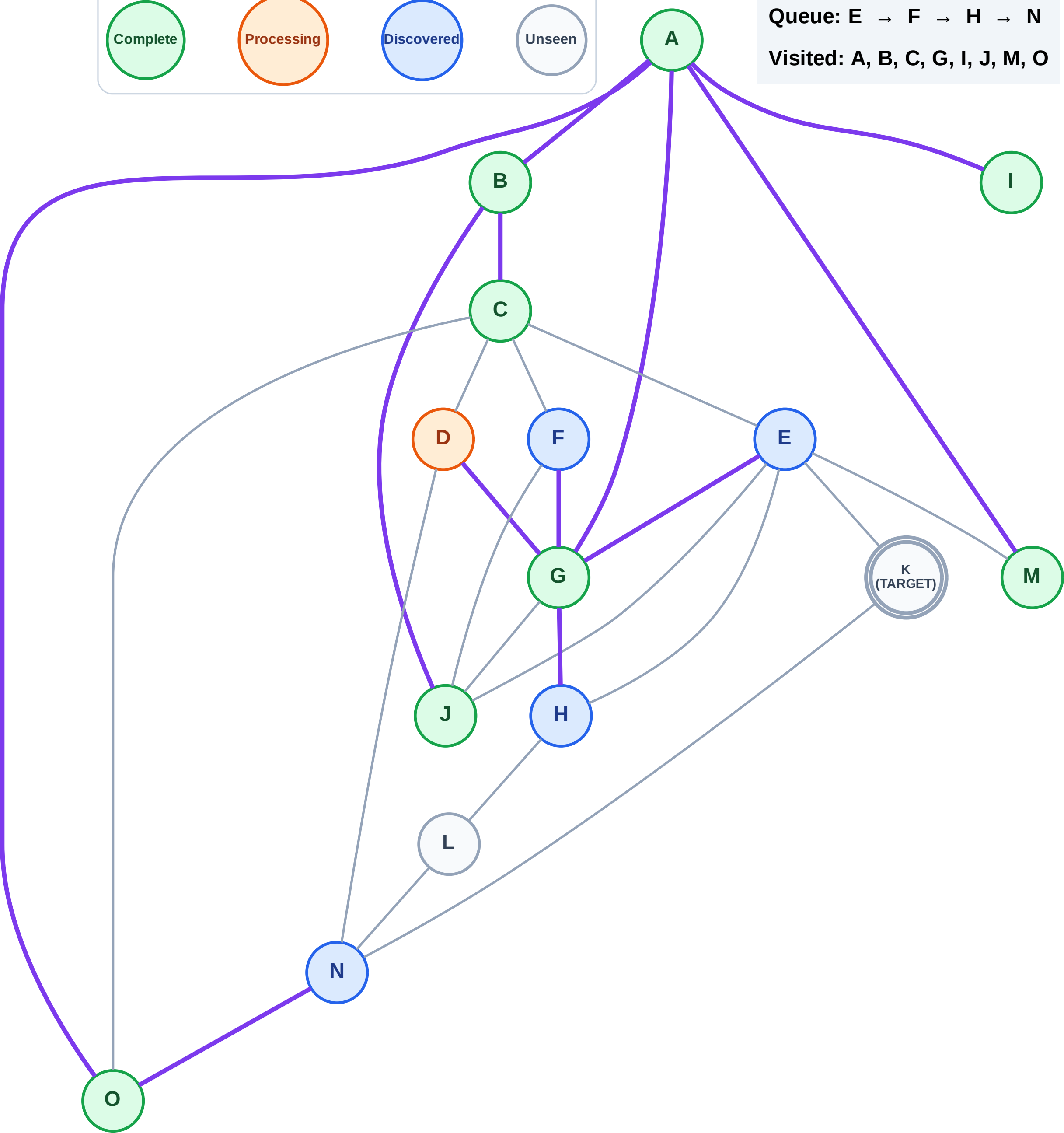
Processing

Discovered

Unseen

Queue: E → F → H → N

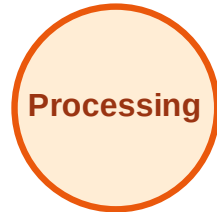
Visited: A, B, C, G, I, J, M, O



BFS Traversal

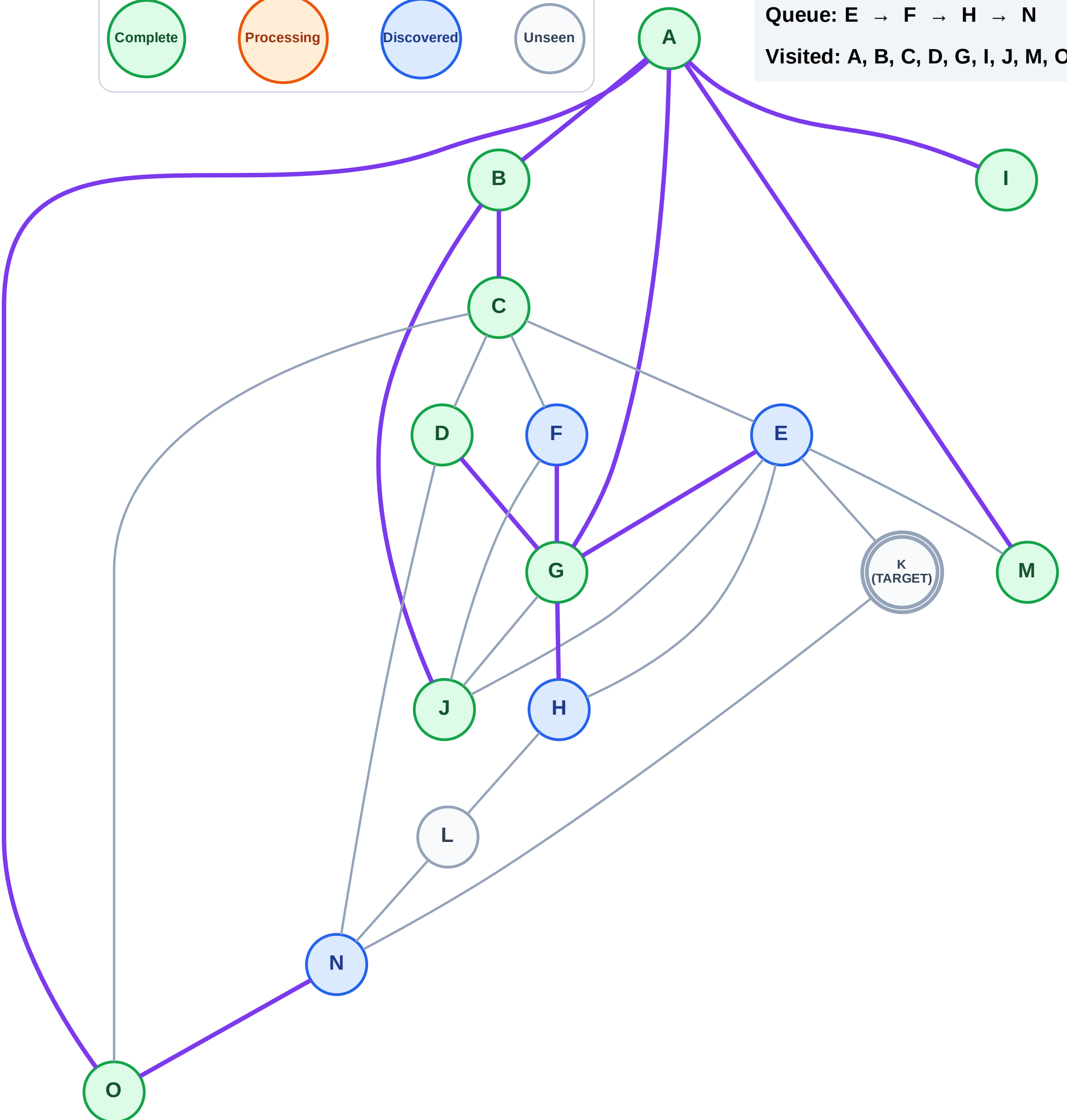
Step 19: No new neighbors from D

Legend



Queue: E → F → H → N

Visited: A, B, C, D, G, I, J, M, O



BFS Traversal

Step 20: Process node E

Legend

Complete

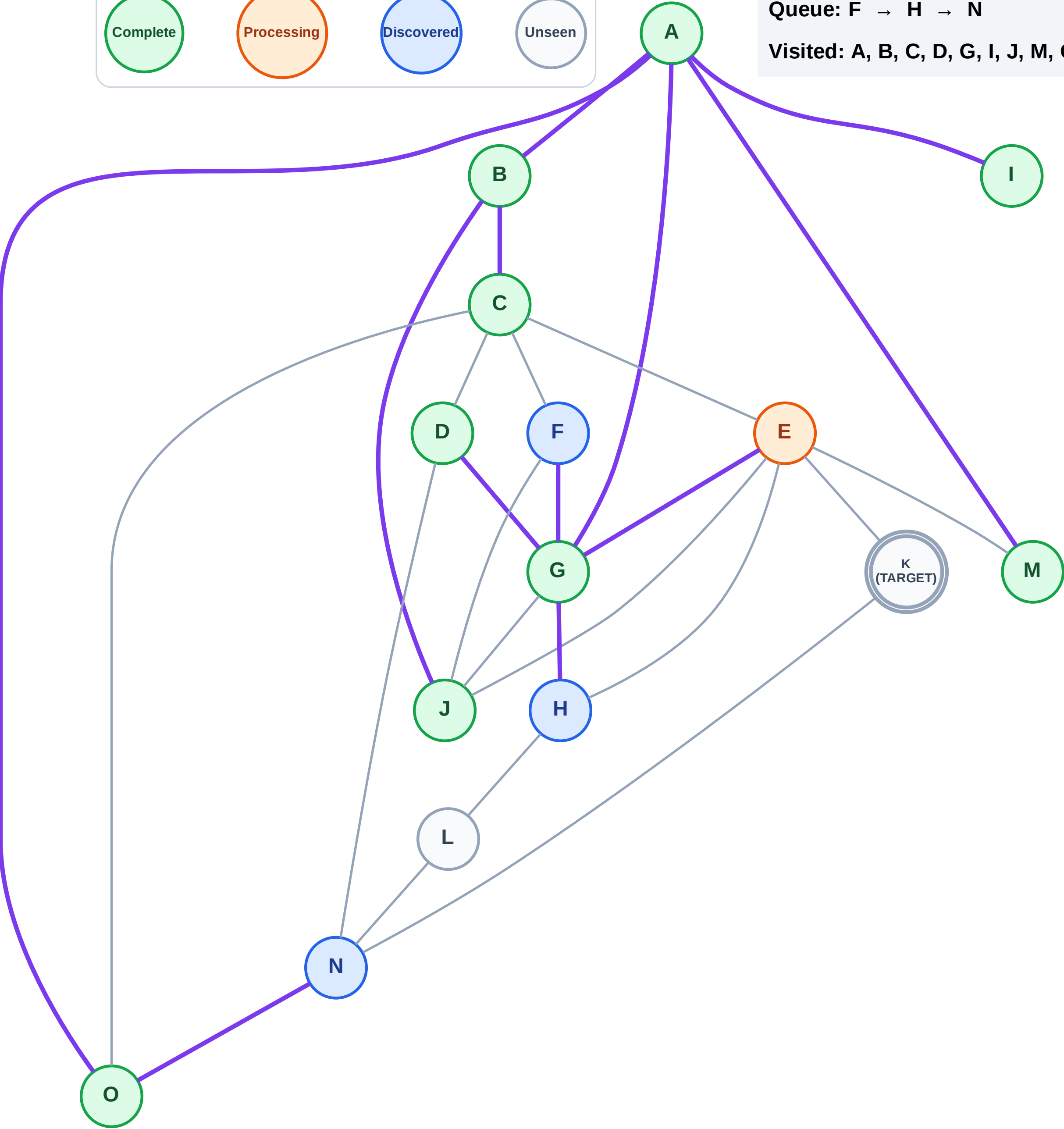
Processing

Discovered

Unseen

Queue: F → H → N

Visited: A, B, C, D, G, I, J, M, O



BFS Traversal

Step 21: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → G → E → K

Legend

Complete

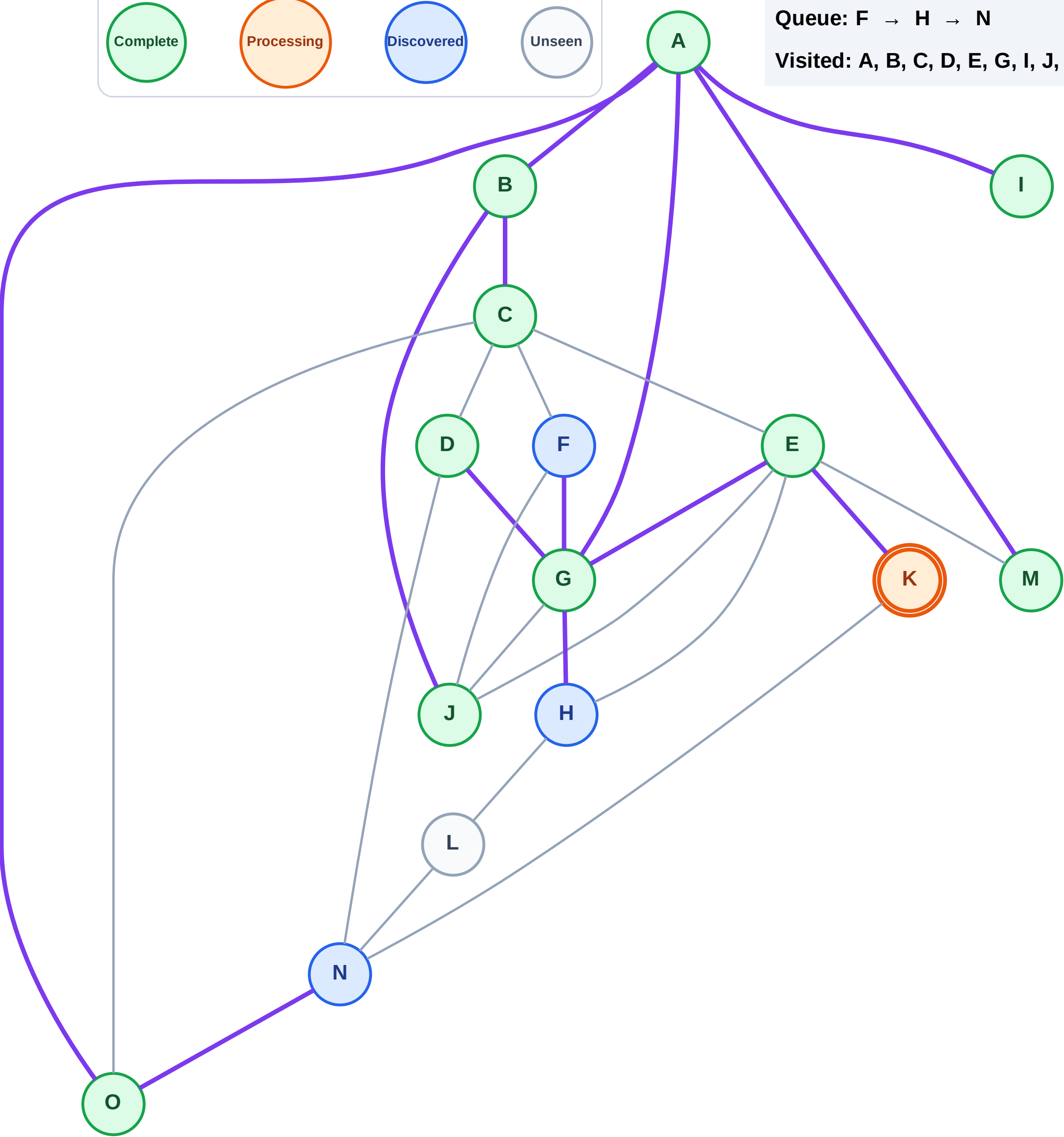
Processing

Discovered

Unseen

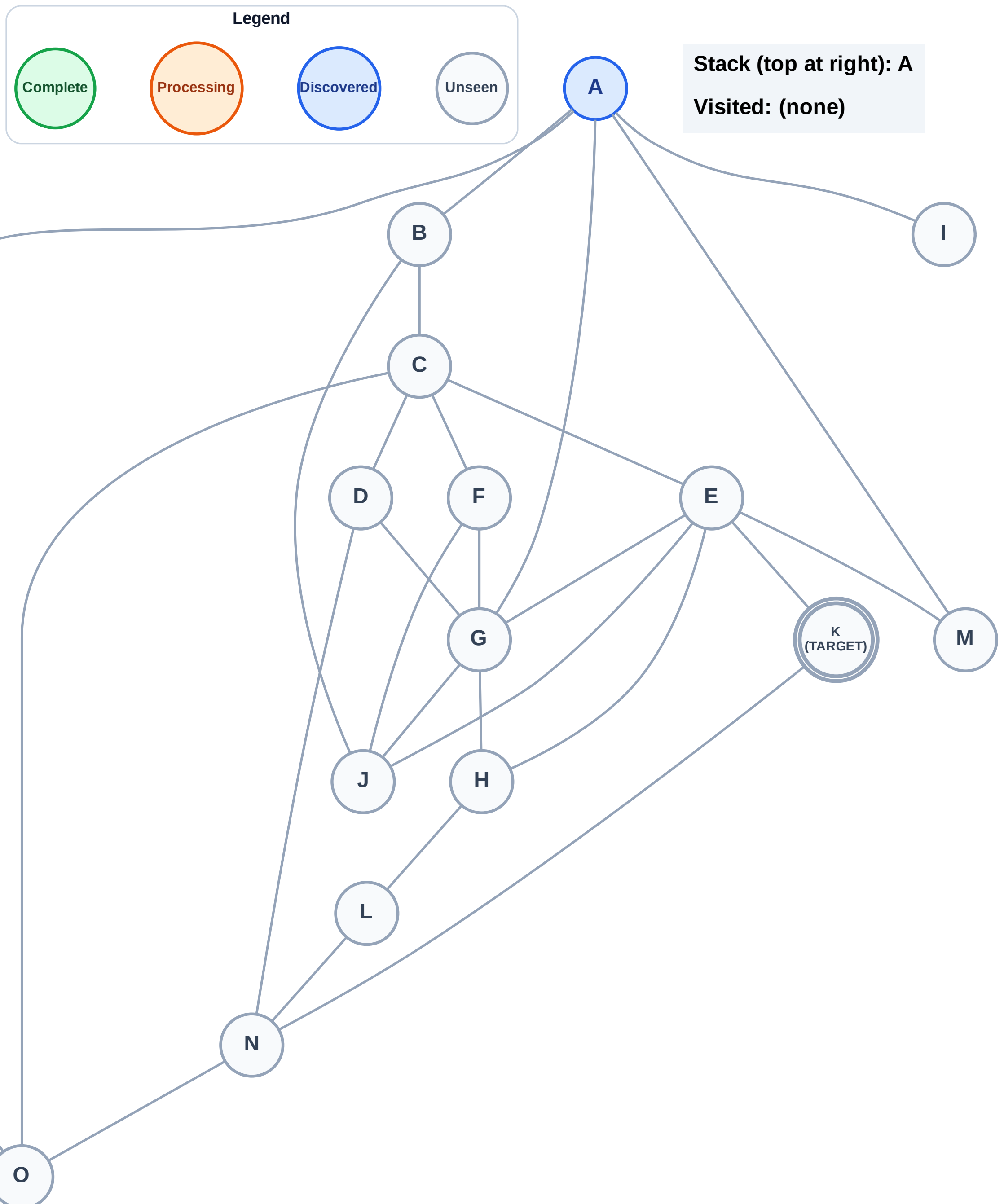
Queue: F → H → N

Visited: A, B, C, D, E, G, I, J, K, M, O



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



DFS Traversal

Step 2: Process node A

Legend

Complete

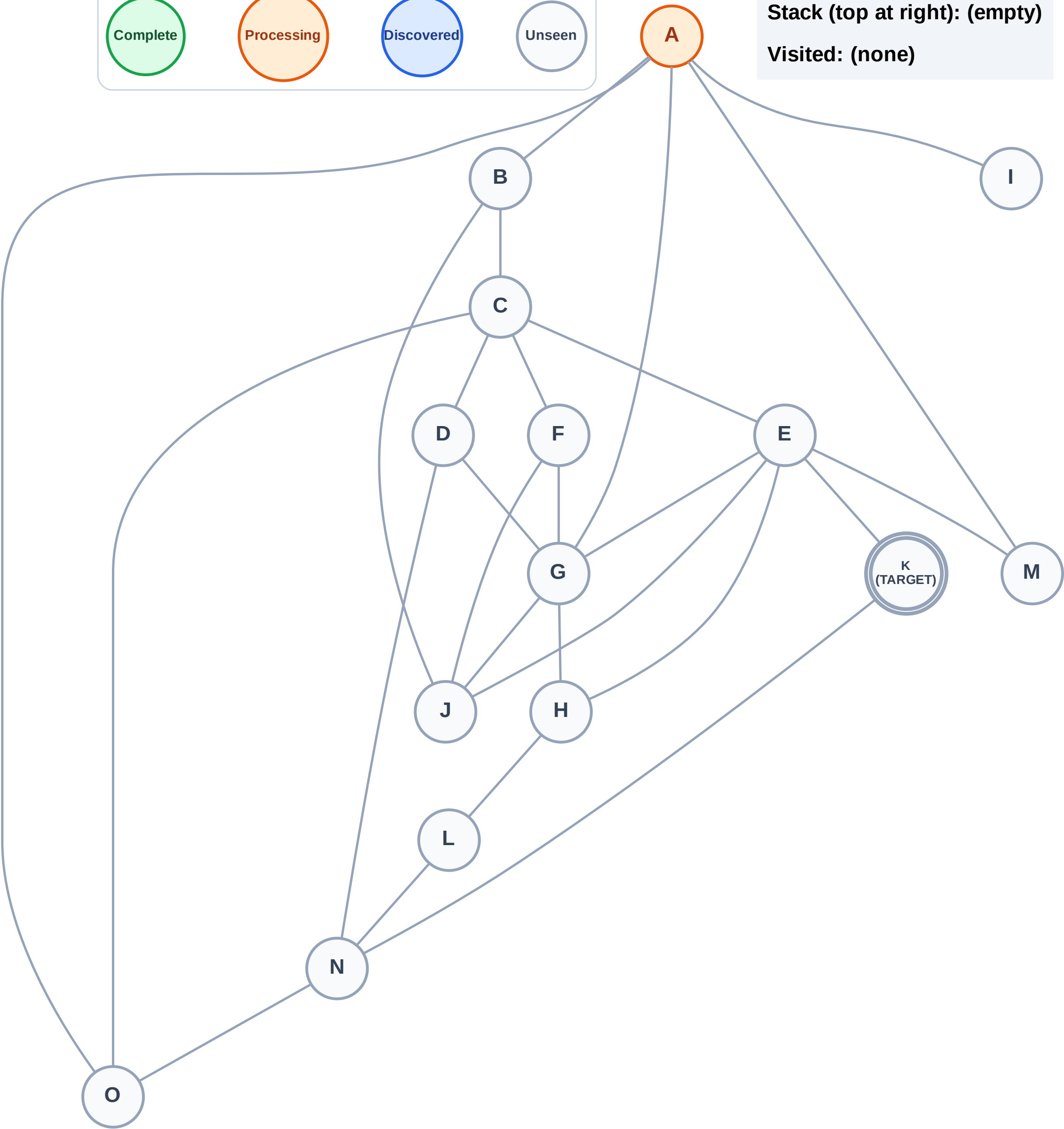
Processing

Discovered

Unseen

Stack (top at right): (empty)

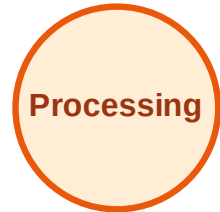
Visited: (none)



DFS Traversal

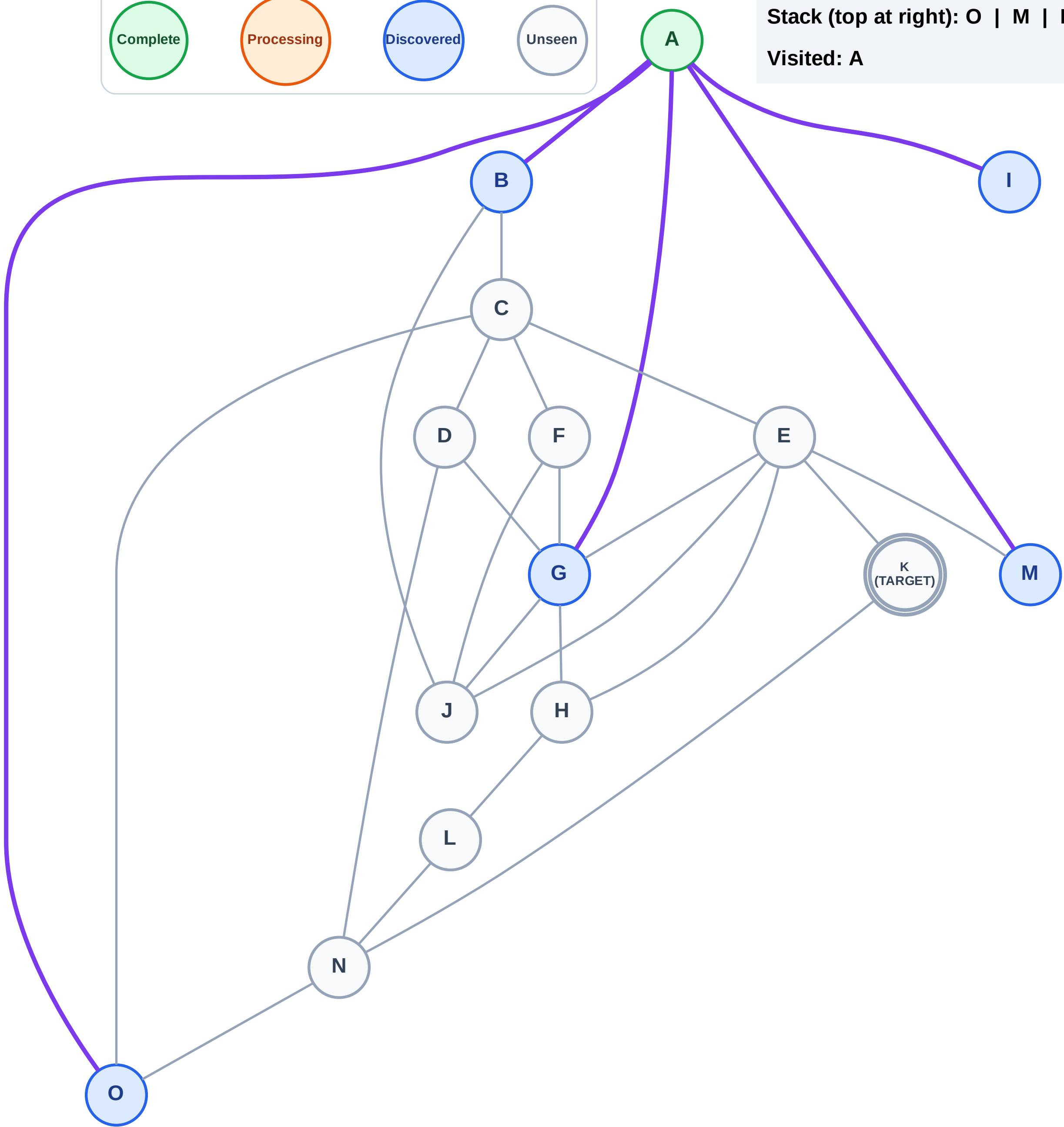
Step 3: Discover O, M, I, G, B from A

Legend



Stack (top at right): O | M | I | G | B

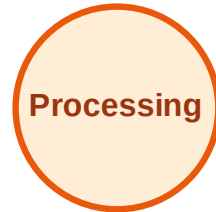
Visited: A



DFS Traversal

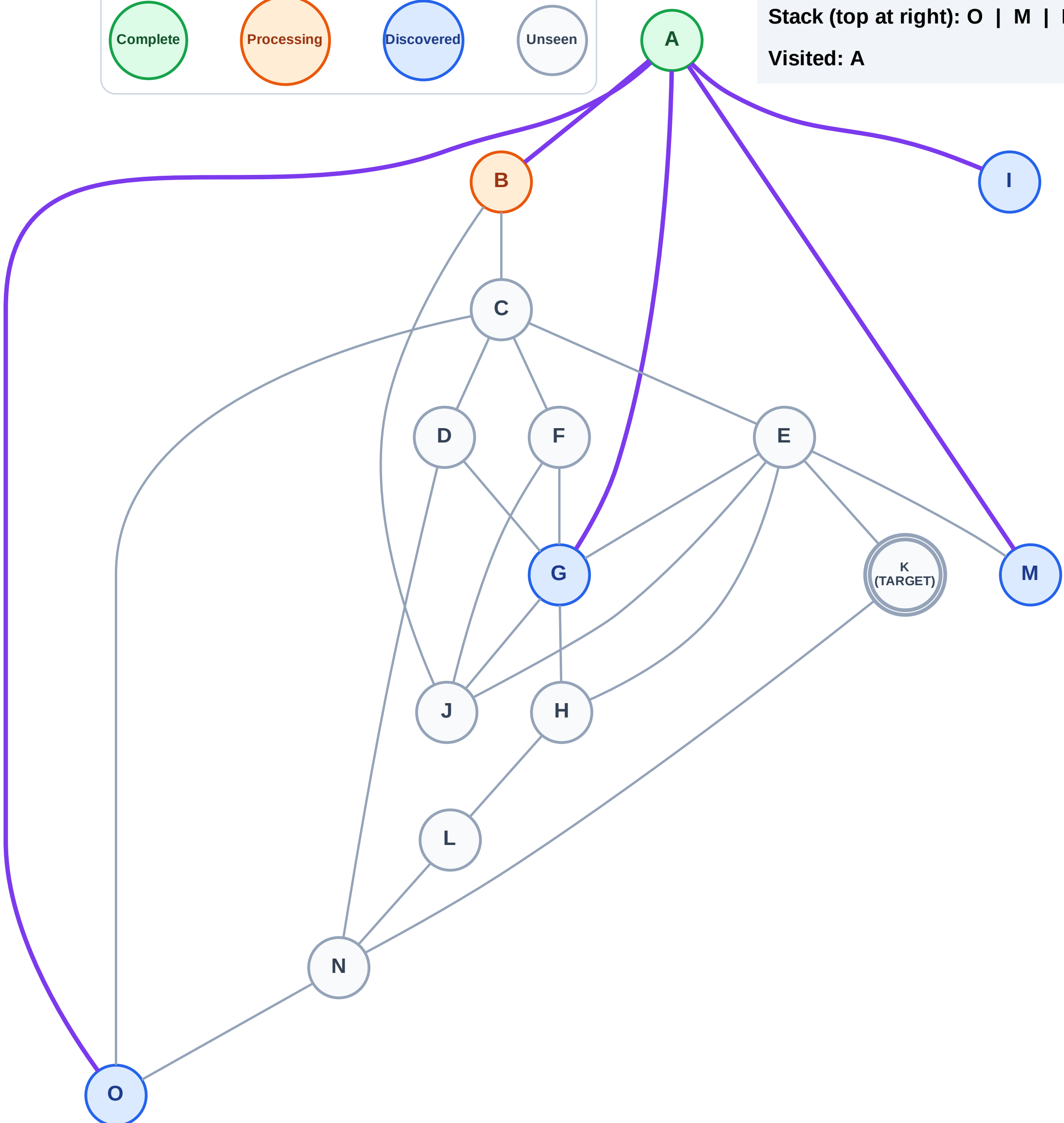
Step 4: Process node B

Legend



Stack (top at right): O | M | I | G

Visited: A



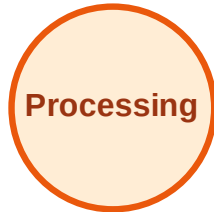
DFS Traversal

Step 5: Discover J, C from B

Legend



Complete



Processing



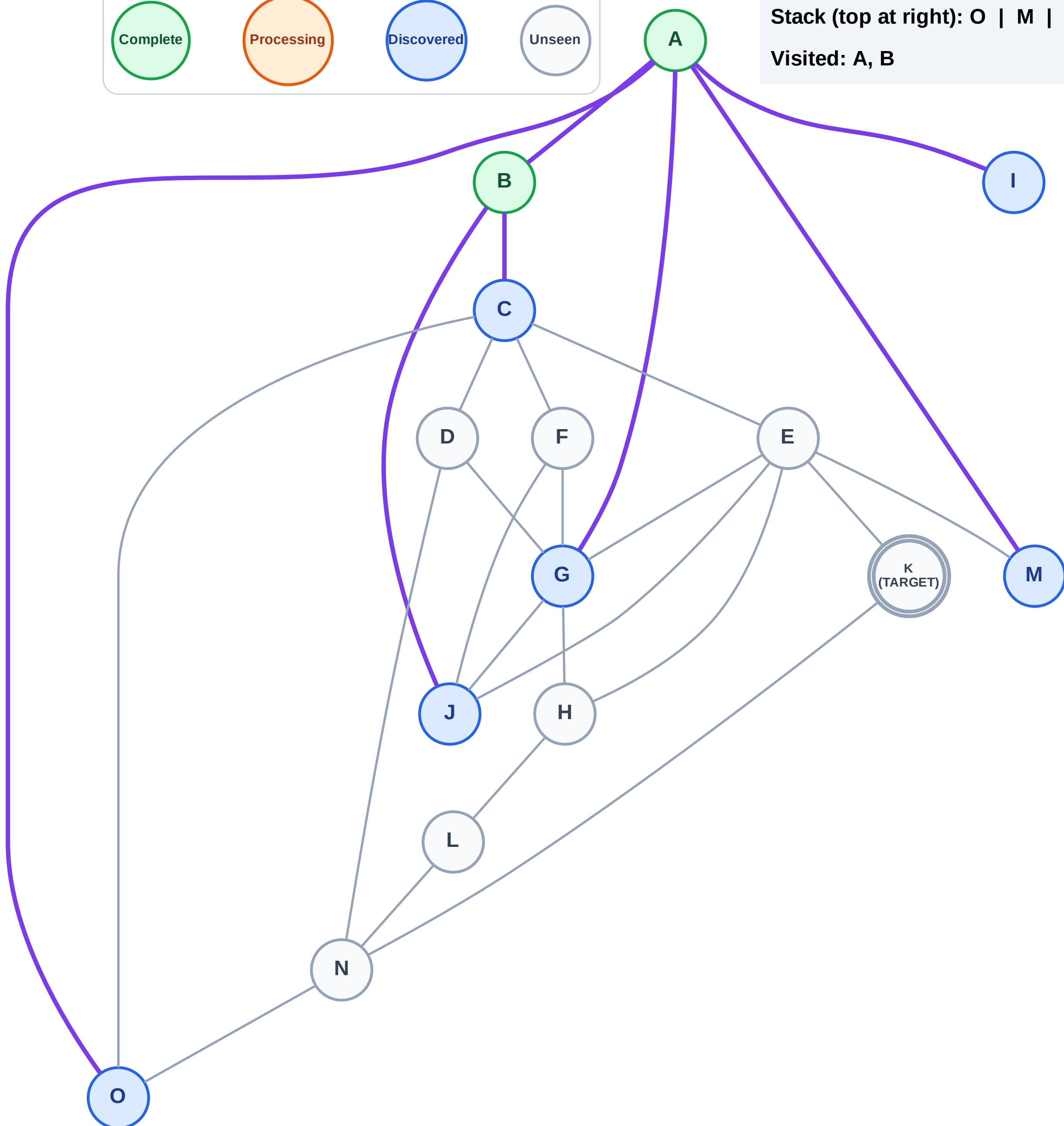
Discovered



Unseen

Stack (top at right): O | M | I | G | J | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

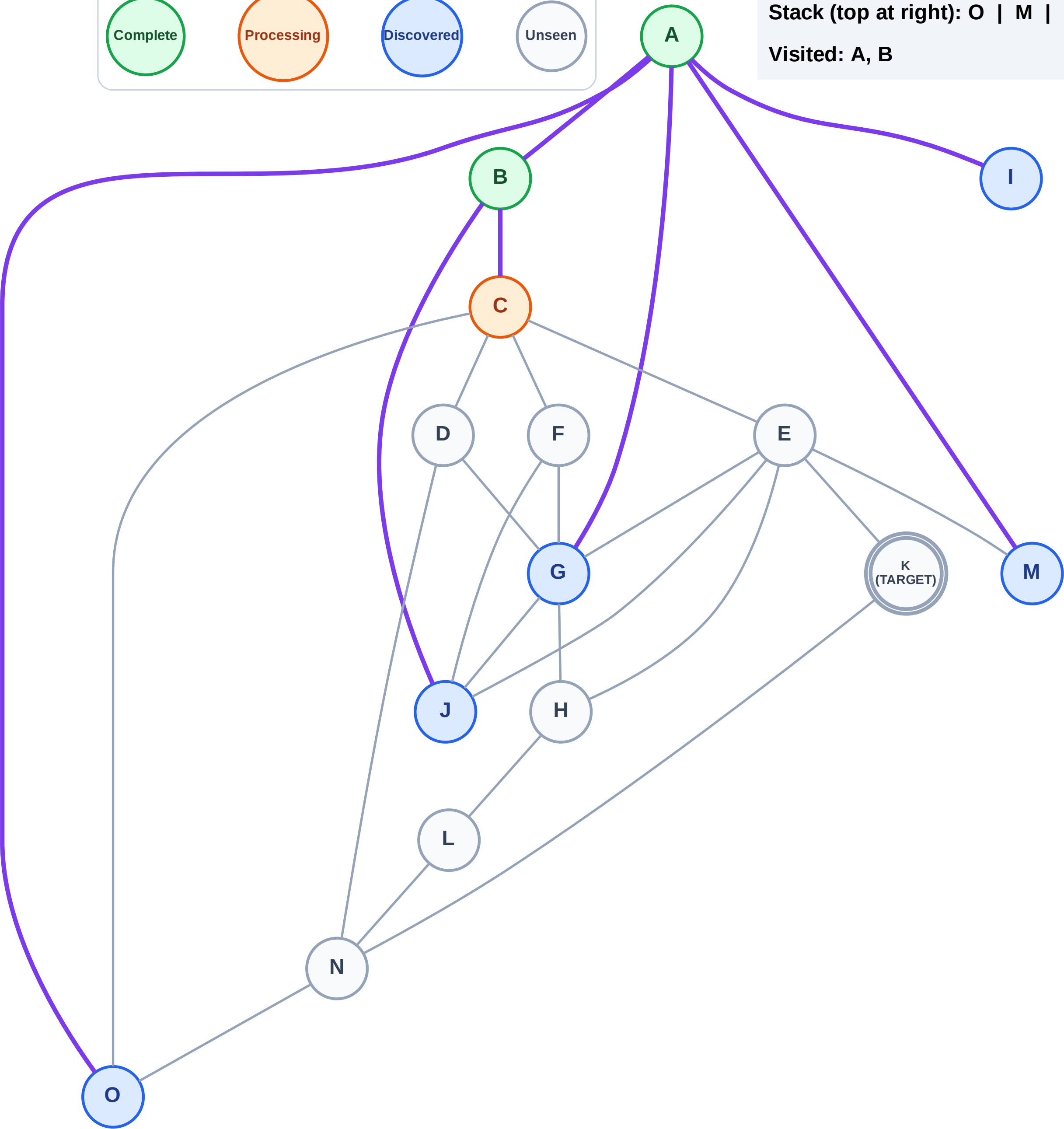
Processing

Discovered

Unseen

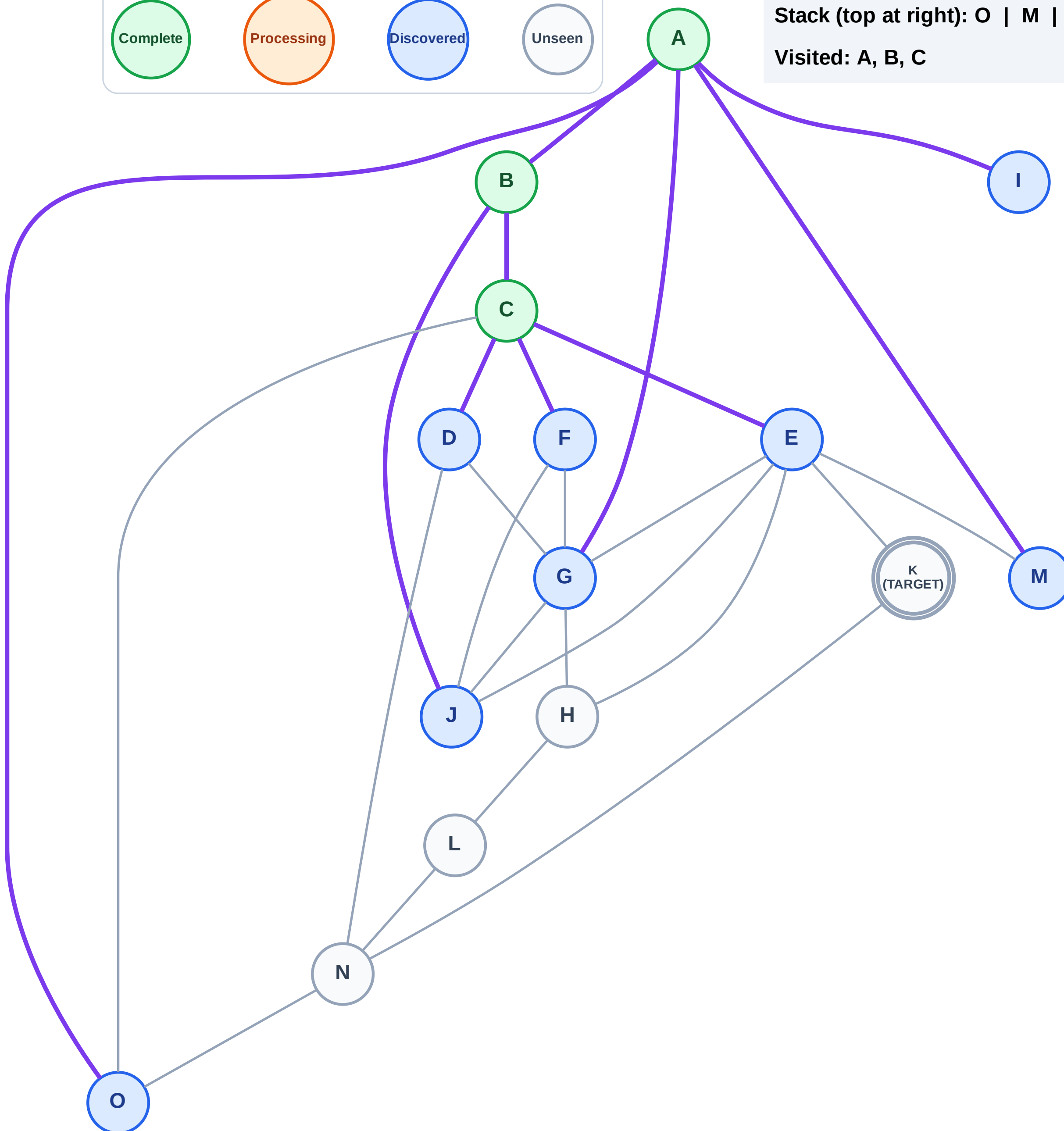
Stack (top at right): O | M | I | G | J

Visited: A, B



Step 7: Discover F, E, D from C

Step 7: Discover F, E, D from C



DFS Traversal

Step 8: Process node D

Legend

Complete

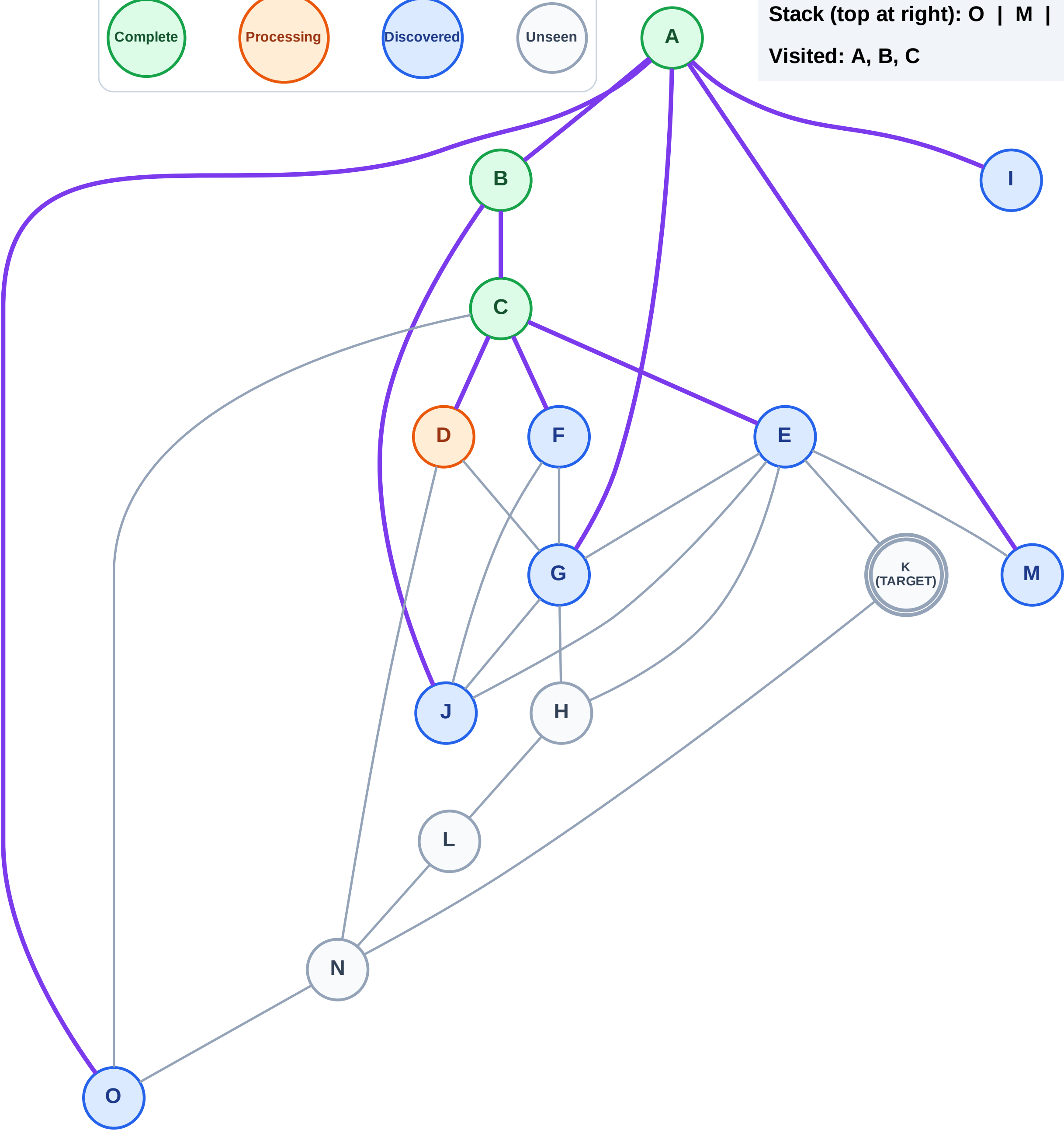
Processing

Discovered

Unseen

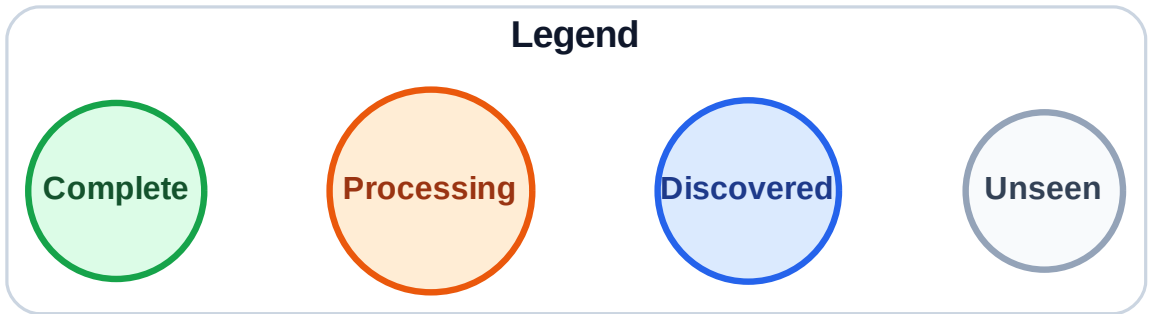
Stack (top at right): O | M | I | G | J | F | E

Visited: A, B, C



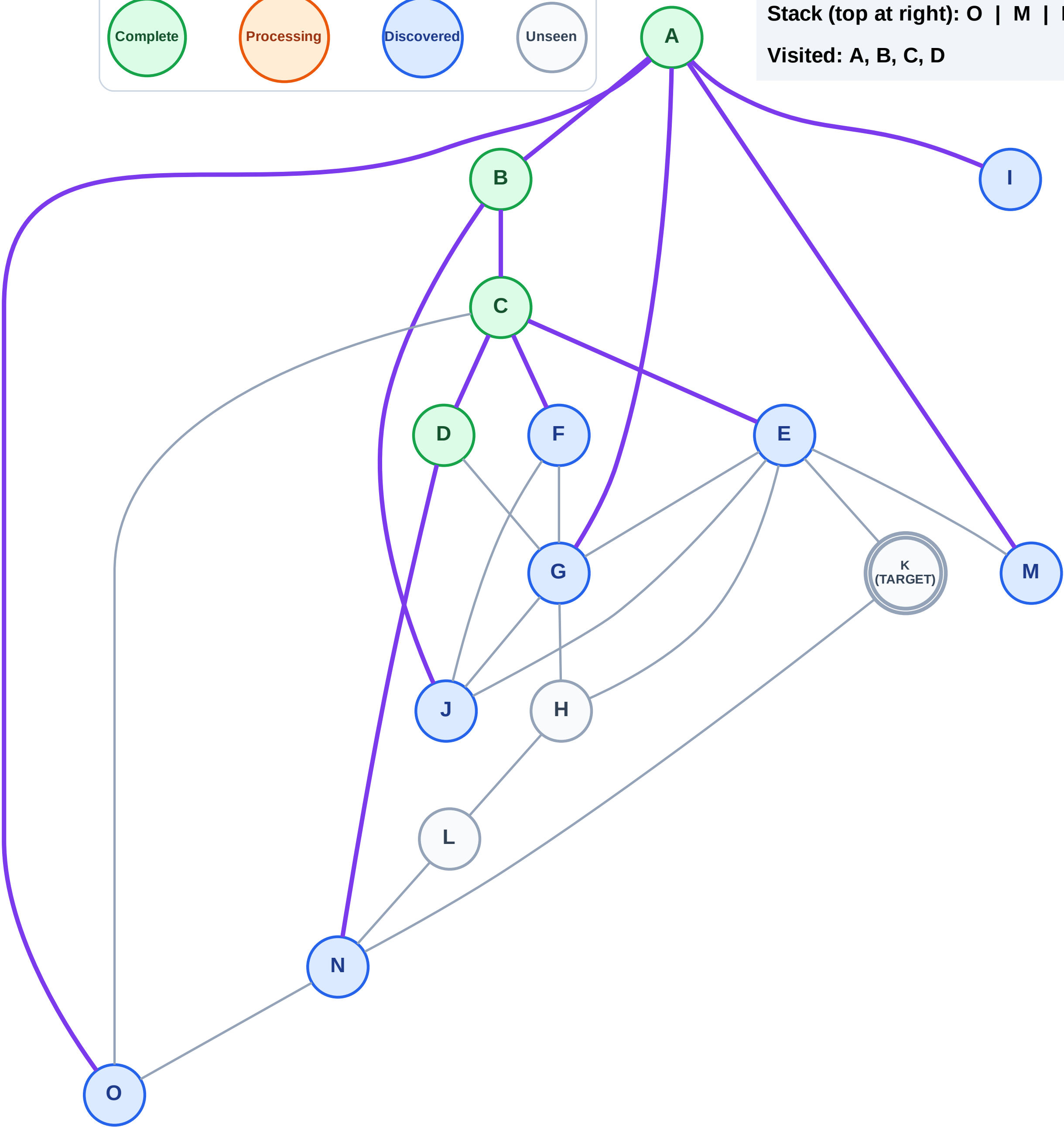
DFS Traversal

Step 9: Discover N from D



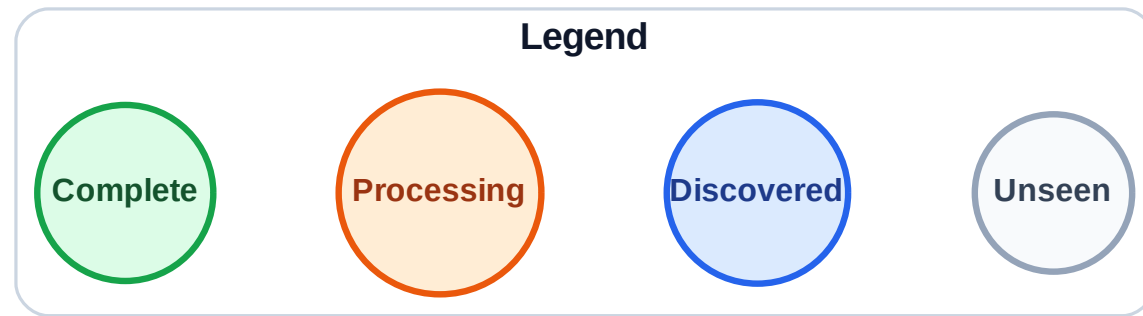
Stack (top at right): O | M | I | G | J | F | E | N

Visited: A, B, C, D

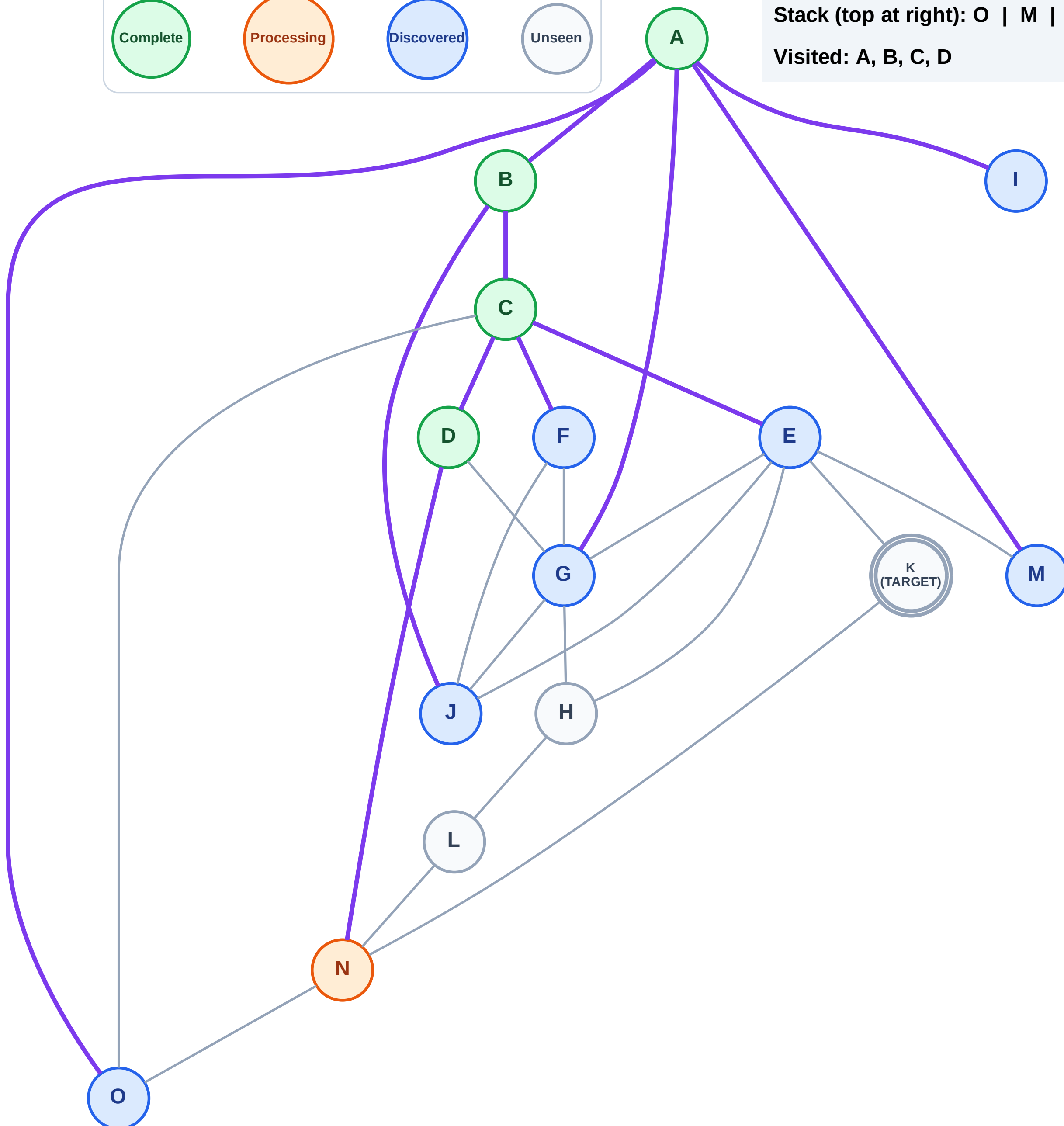


DFS Traversal

Step 10: Process node N



Stack (top at right): O | M | I | G | J | F | E
Visited: A, B, C, D



DFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (5 hops): A → B → C → D → N → K

Legend



Stack (top at right): O | M | I | G | J | F | E | L

Visited: A, B, C, D, K, N

